

---

# Confidence Calibration under Ambiguous Ground Truth

---

Anonymous Author(s)

Affiliation

Address

email

## Abstract

1 Confidence calibration implicitly assumes a unique ground-truth label per input.  
2 This assumption breaks down in settings such as medical diagnosis, natural lan-  
3 guage inference, and perceptual tasks, where multiple annotators may genuinely  
4 disagree. We show that post-hoc calibrators fitted on majority-voted labels can  
5 appear well-calibrated against voted labels yet remain substantially miscalibrated  
6 against labels drawn from the underlying ambiguous label distribution reflected  
7 by human disagreement. This failure is structural, affecting classic calibration  
8 methods such as Temperature Scaling, Platt scaling, and Histogram Binning. We  
9 prove that Temperature Scaling is biased toward lower temperatures than this  
10 ambiguity-aware target distribution requires, and that the resulting true-label error  
11 increases with annotation entropy. To address this failure, we propose a family of  
12 ambiguity-aware post-hoc calibrators that optimize proper scoring rules against  
13 the full label distribution, among which **Dirichlet-Soft** — a full  $K \times K$  affine  
14 calibrator fitted against the annotator distribution — consistently achieves the best  
15 distributional fit. For the more common case where the calibration set contains only  
16 one-hot voted labels, we propose **Label-Smooth Temperature Scaling (LS-TS)**,  
17 which constructs pseudo-soft targets from the model’s own confidence without any  
18 additional annotations. Experiments on CIFAR-10H, ChaosNLI, ISIC 2019, and  
19 DermaMNIST, spanning eight architectures across vision and language, show that  
20 Dirichlet-Soft reduces true-label ECE by 59–86% relative to standard Temperature  
21 Scaling and achieves the best Brier score and NLL across all benchmarks, while  
22 LS-TS reduces true-label ECE by 7–78% without requiring annotator distributions.

## 23 1 Introduction

24 A well-calibrated classifier  $f : \mathcal{X} \rightarrow \Delta^K$  produces probability vectors  $\hat{p}(x) = f(x)$  such that stated  
25 confidence matches empirical accuracy: when  $\hat{p}_{\hat{c}}(x) = p$  for the predicted class  $\hat{c} = \arg \max_k \hat{p}_k(x)$ ,  
26 the true probability of  $\hat{c}$  being correct is exactly  $p$ . Formally,

$$\mathbb{P}(\hat{Y} = Y \mid \hat{p}_{\hat{Y}}(X) = p) = p, \quad \forall p \in [0, 1], \quad (1)$$

27 where  $\Delta^K$  denotes the  $K$ -dimensional probability simplex. Calibration is critical for safe deployment  
28 in medicine [Kompa et al., 2021], autonomous driving, and other high-stakes settings.

29 **The hidden assumption.** Equation (1) implicitly assumes a unique ground-truth label  $Y$  for every  
30 input  $x$ . In practice this assumption frequently fails: a skin lesion may be legitimately classified as  
31 malignant or benign by different dermatologists; a natural-language premise may or may not entail a  
32 hypothesis depending on pragmatic context; low-resolution objects can be categorically ambiguous.  
33 In such settings the correct target is not a single label but a *distribution over labels*,  $\pi(\cdot \mid x) \in \Delta^K$ ,  
34 representing the underlying ambiguous label distribution induced by genuine annotator disagreement.

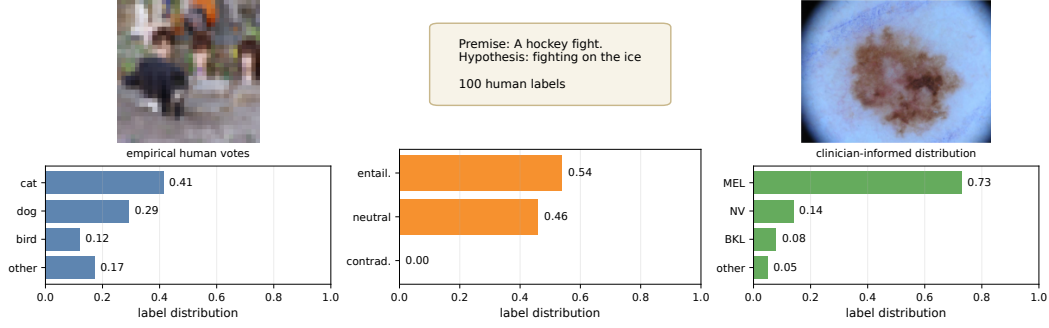


Figure 1: **Left:** a CIFAR-10H image with dispersed human votes (cat/dog/bird), illustrating perceptual ambiguity in low-resolution vision. **Middle:** a ChaosNLI premise–hypothesis pair with split entailment/neutral judgments, illustrating semantic ambiguity. **Right:** an ISIC 2019 melanoma image paired with the clinician-informed label distribution used in our medical experiments, showing clinically plausible MEL/NV confusion. CIFAR-10H and ChaosNLI distributions are empirical human label distributions; the ISIC panel uses the dermatologist confusion model described in Appendix I.

35 **The standard practice and its failure.** Figure 1 illustrates the core issue: a single input can  
 36 support multiple plausible labels, and the full label distribution is the object that should be calibrated.  
 37 Standard practice nevertheless aggregates annotations by majority vote to obtain a *voted label*  $y^*$  and  
 38 calibrates against it, and in many applications this one-hot voted label is all that is available in the  
 39 calibration set. We show that this can yield low ECE against voted labels ( $\text{ECE}_{\text{voted}}$ ) while leaving  
 40 substantial miscalibration against true labels drawn from the ambiguity-aware label distribution  
 41  $\pi(\cdot | x)$  ( $\text{ECE}_{\text{true}}$ , following Stutz et al. [2023]). The mechanism is straightforward: ambiguous  
 42 examples appear underconfident relative to the one-hot voted target, so Temperature Scaling selects  
 43 a lower  $T$  than the ambiguity-aware label distribution requires, amplifying overconfidence. In a  
 44 controlled experiment (Section 4), Temperature Scaling, Platt scaling, and Histogram Binning all fail  
 45 to correct this mismatch, because the root cause is the voted-label target, not method capacity.

## 46 Contributions.

- 47 1. **Problem and theory** (Sections 3–5): we formalize true-label calibration under ambiguous  
 48 labels and show that TS is biased toward overly low temperatures, with true-label miscalibration  
 49 becoming more severe as annotation entropy increases.
- 50 2. **Methods** (Section 6): we propose ambiguity-aware post-hoc calibrators — notably **Dirichlet-Soft**,  
 51 which consistently achieves the best distributional fit — for settings with annotator distributions,  
 52 and **LS-TS** for the more realistic setting where the calibration set contains only one-hot voted  
 53 labels.
- 54 3. **Evidence** (Sections 4 and 7): a controlled toy example and four real benchmarks show that  
 55 standard calibrators can worsen true-label calibration, while Dirichlet-Soft reduces  $\text{ECE}_{\text{true}}$  by  
 56 59–86% and LS-TS by 7–78% consistently across vision and language.

## 57 2 Related Work

58 **Confidence calibration.** Guo et al. [2017] showed that modern networks are miscalibrated and  
 59 that TS is an effective post-hoc remedy. Isotonic regression [Zadrozny and Elkan, 2002], Platt  
 60 scaling [Platt, 1999], and Dirichlet calibration [Kull et al., 2019] extend this to multiclass settings,  
 61 all assuming a unique ground-truth label. Minderer et al. [2021] showed ViTs are better calibrated  
 62 than CNNs; Bai et al. [2021] revisit the sufficiency of TS. All share the voted-label assumption;  
 63 we show this assumption, not method capacity, is the limiting factor. Label smoothing [Szegedy  
 64 et al., 2016, Müller et al., 2019] applies uniform smoothed targets during training; we instead use  
 65 annotation-informed label distributions for post-hoc calibration requiring no retraining.

66 **Ambiguous ground truth.** Dense annotation datasets [Peterson et al., 2019, Nie et al., 2020] and  
 67 multi-annotator learning [Uma et al., 2021, Rodrigues and Pereira, 2018] have received growing

attention. Baan et al. [2022] show ECE is theoretically ill-defined when humans genuinely disagree and propose instance-level alternatives; our work complements this by providing methods that improve true-label calibration under that setting. Gordon et al. [2022] and Plank [2022] highlight epistemological problems with annotation aggregation in NLP. Khurana et al. [2024] use disagreement as an abstention signal; Singh et al. [2025] show training on soft labels preserves epistemic uncertainty, a training-time complement to our post-hoc approach.

**Conformal prediction under ambiguous ground truth.** Stutz et al. [2023] show that conformal sets calibrated on voted labels undercover the true annotator distribution, the analogous failure for set prediction that our work addresses for point prediction. Calibration under covariate shift [Park et al., 2020, Wang et al., 2020] is orthogonal to our label-ambiguity setting.

### 3 Problem Formulation

#### 3.1 Setup and Notation

Let  $\mathcal{Y} = \{1, \dots, K\}$  be the label space. A classifier  $f : \mathcal{X} \rightarrow \Delta^K$  outputs  $\hat{p}(x) = \text{softmax}(z(x))$  from pre-softmax logits  $z(x) \in \mathbb{R}^K$ . The **annotator distribution**  $\pi(\cdot | x) \in \Delta^K$  denotes the underlying ambiguous label distribution for input  $x$  and is estimated from  $m$  annotations as  $\hat{\pi}_k(x) = \frac{1}{m} \sum_j \mathbf{1}[a_j = k]$ . The **voted label** is  $y^* = \arg \max_k \hat{\pi}_k(x)$ . Standard calibration uses one-hot voted labels. For example, Temperature Scaling [Guo et al., 2017] minimizes

$$\mathcal{L}_{\text{TS}}(T) = -\frac{1}{n} \sum_{i=1}^n \log \text{softmax}(z_i/T)_{y_i^*}. \quad (2)$$

#### 3.2 True-Label Calibration

Following Stutz et al. [2023], we distinguish the voted label  $y^*$  from the **true label**  $\tilde{y} \sim \pi(\cdot | x)$ , drawn from the underlying ambiguous label distribution.

**Definition 1** (True-Label Calibration).  *$f$  is **true-label calibrated** if for all  $p \in [0, 1]$ :  $\mathbb{P}(\tilde{Y} = \hat{c}(X) | \hat{p}_{\hat{c}}(X) = p) = p$ , where  $\hat{c}(x) = \arg \max_k \hat{p}_k(x)$  and  $\tilde{Y} \sim \pi(\cdot | X)$ .*

The **voted-label ECE**  $\text{ECE}_{\text{voted}}$  is the standard ECE against  $y^*$ . The **true-label ECE**  $\text{ECE}_{\text{true}}$  averages ECE over 100 draws  $\tilde{y}_i \sim \pi(\cdot | x_i)$  ( $B = 15$  equal-width bins). Throughout the paper we compare these two views directly, since ambiguous examples can look well-calibrated under voted labels while remaining overconfident under the underlying ambiguous label distribution.

### 4 Motivating Example

**Setup.** We construct a 3-class 2D Gaussian dataset with three clusters:  $x \sim \mathcal{N}((-3.2, 1.1), \text{diag}(0.60, 0.45))$  for class 0 with  $\pi = [1, 0, 0]$ ;  $x \sim \mathcal{N}((0, 0), \text{diag}(1.15, 0.75))$  for the ambiguous cluster with  $\pi = [0, 0.70, 0.30]$ ; and  $x \sim \mathcal{N}((3.2, -1.1), \text{diag}(0.60, 0.45))$  for class 2 with  $\pi = [0, 0, 1]$ . The voted label for the middle cluster is always class 1 because  $0.70 > 0.30$ . A 2-layer MLP (64 hidden units per layer) is trained on voted labels, and all calibrators are fitted on a held-out calibration split using the same voted labels.

**Results.** Figure 2 shows that this mismatch is a structural consequence of the voted-label target. TS reduces  $\text{ECE}_{\text{voted}}$  from 3.25% to 1.11%, but *increases*  $\text{ECE}_{\text{true}}$  from 6.26% to 9.64%. Platt scaling behaves similarly ( $\text{ECE}_{\text{voted}} = 0.68\%$ ,  $\text{ECE}_{\text{true}} = 9.70\%$ ), and Histogram Binning also lowers voted-label ECE while worsening true-label ECE (1.32%/9.65%). Figure 2(f) further shows that this residual error is concentrated in the ambiguous cluster for all three traditional methods. TS sets  $T = 0.62 < 1$ , boosting confidence toward the voted target and thereby pushing the model *further* from the true ambiguous label distribution. The failure is therefore not specific to TS: standard post-hoc methods are useful for  $\text{ECE}_{\text{voted}}$ , but the voted-label target makes them ineffective for  $\text{ECE}_{\text{true}}$ .

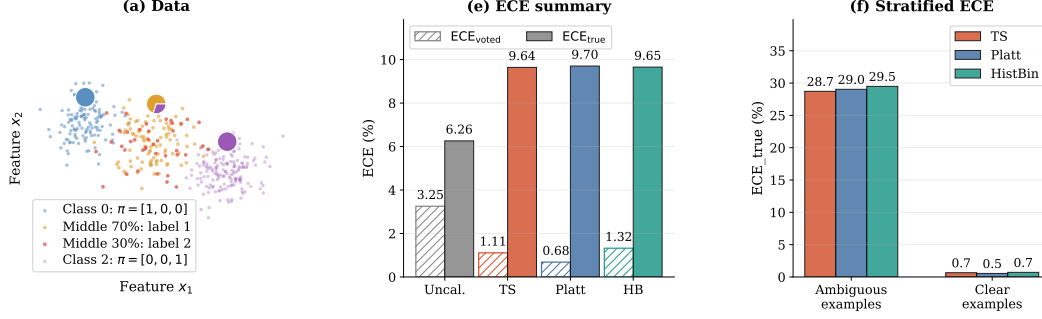


Figure 2: **Motivating example: all standard calibration methods fail.** (a) Toy dataset generation: only the middle Gaussian cluster is ambiguous, with  $\pi = [0, 0.70, 0.30]$ ; orange points are drawn as label 1 (70%) and red points as label 2 (30%), while all receive the same voted label 1. (e) Summary of the toy results: every traditional calibrator lowers  $\text{ECE}_{\text{voted}}$  but *increases*  $\text{ECE}_{\text{true}}$ , so voted-label evaluation masks the failure. (f) Stratified  $\text{ECE}_{\text{true}}$  for TS, Platt scaling, and Histogram Binning: the residual error is concentrated in ambiguous examples for all three traditional methods.

## 5 Theory: Why Standard Calibration Fails

We now formalize two complementary aspects of the failure of voted-label calibration under ambiguity: the *direction* of the TS bias (Proposition 1) and its *scaling* with label ambiguity (Proposition 2).

**Proposition 1** (Direction of TS Bias). *Consider a calibration set containing an ambiguous cluster in which the voted label  $y^*$  equals the majority class for every example, but the true annotator probability of  $y^*$  is  $\pi_{y^*}(x) = q < 1$ . If the model is already reasonably accurate on this cluster ( $\text{softmax}(z_i)_{y^*} > q$ ), then TS minimizing  $\mathcal{L}_{\text{TS}}$  finds  $T^* < 1$  (increasing model confidence), whereas ambiguity-aware calibration requires  $T^* > 1$  (reducing confidence to  $q$ ).*

*Proof sketch.* On the ambiguous cluster, all voted labels are  $y^*$ , so the voted-label loss (2) is minimized by pushing  $\text{softmax}(z_i/T)_{y^*} \rightarrow 1$ , i.e.  $T \rightarrow 0$ . Hence  $\partial \mathcal{L}_{\text{TS}} / \partial T > 0$  and the optimal  $T^* < 1$ . Ambiguity-aware calibration instead minimizes  $\mathcal{L}_{\text{SLTS}}$  with target  $q < 1$ ; the minimum requires  $\text{softmax}(z_i/T^*)_{y^*} \approx q$ , necessitating  $T^* > 1$  whenever  $\text{softmax}(z_i)_{y^*} > q$ . Full proof in Appendix B.  $\square$

**Proposition 2** (True-Label Miscalibration and Annotation Entropy). *Let  $H(x) = -\sum_k \pi_k(x) \log \pi_k(x)$  denote the annotation entropy of  $x$ . Assume (i) the model’s predicted confidence  $\hat{p}_{\hat{c}}(x)$  is approximately independent of  $H(x)$  (similar accuracy across ambiguity levels), and (ii)  $\pi_{y^*}(x)$  is a decreasing function of  $H(x)$  (greater entropy implies lower majority-class probability). Then the expected true-label calibration error of voted-label-calibrated predictions is non-decreasing in  $H(x)$ .*

*Proof sketch.* For unambiguous examples ( $H(x) = 0$ ),  $\pi(\cdot | x)$  is one-hot, so true-label and voted-label evaluation coincide. As  $H(x)$  increases,  $\pi_{y^*}(x)$  decreases below 1. The true-label calibration error for the predicted class is  $|\hat{p}_{\hat{c}}(x) - \pi_{\hat{c}}(x)|$ ; since the model is trained against the voted label (targeting  $\approx 1$ ),  $\hat{p}_{\hat{c}}(x) > \pi_{y^*}(x)$ , and this mismatch grows as  $\pi_{y^*}(x)$  falls. Hence the true-label calibration error increases with  $H(x)$ .  $\square$

## 6 Methods

All methods in this paper are *post-hoc*: they operate on cached logits from a fixed pre-trained model and require only a calibration set equipped with annotator distributions or individual annotations. No retraining or architectural modification is needed. Training-time approaches such as soft-label training [Szegedy et al., 2016, Müller et al., 2019], mixup [Thulasidasan et al., 2019], and multi-annotator learning [Uma et al., 2021, Rodrigues and Pereira, 2018] can also improve calibration under ambiguity, but they require access to the training pipeline and are orthogonal to post-hoc calibration; we do not compare against them here.

## 6.1 Ambiguity-Aware Calibration Methods

**Soft-Label Temperature Scaling (SLTS).** SLTS minimizes the cross-entropy between the calibrated output and the empirical label distribution:

$$\begin{aligned}\mathcal{L}_{\text{SLTS}}(T) &= -\frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K \hat{\pi}_k(x_i) \log \text{softmax}(z_i/T)_k \\ &= \frac{1}{n} \sum_{i=1}^n \text{KL}(\hat{\pi}(x_i) \parallel \text{softmax}(z_i/T)) + \text{const.}\end{aligned}\quad (3)$$

SLTS uses the same single-parameter family as TS but with a target that correctly reflects the annotator distribution.

**Proposition 3** (Proper scoring rule).  *$\mathcal{L}_{\text{SLTS}}$  is a proper scoring rule with respect to  $\hat{\pi}$ : it is uniquely minimized (over all distributions, not just the one-parameter TS family) when  $\text{softmax}(z/T) = \hat{\pi}(x)$  for every  $x$ .*

**Vector Scaling (VS).** Extends SLTS with per-class temperatures  $\mathbf{T} = (T_1, \dots, T_K)$ , accommodating datasets where some classes are systematically more ambiguous than others.

**Distributional Isotonic Regression (IR-Soft).** Fits a monotone mapping from predicted confidence to mean annotator accuracy via the Pool Adjacent Violators Algorithm (PAVA), using the empirical label distribution as target.

**SoftPlatt.** The diagonal affine transform  $W \cdot z + b$  (same structure as Platt) fitted against the annotator distribution via KL divergence. This isolates the contribution of the distributional target from any architectural difference.

**Dirichlet-Soft.** The full  $K \times K$  affine  $Wz + b$  fitted by minimizing  $\text{KL}(\hat{\pi}(x) \parallel \text{softmax}(Wz + b))$  with ODIR regularization ( $\lambda = 10^{-3}$ ) [Kull et al., 2019]. This is the most expressive parametric ambiguity-aware calibrator.

## 6.2 Annotation-Free Calibration

The methods above require annotator distributions  $\hat{\pi}(x_i)$  at calibration time. When only voted labels  $y_i^*$  are available, we propose **Label-Smooth Temperature Scaling (LS-TS)**, which constructs a pseudo-target distribution from the model’s own predictions without any additional annotations.

Let  $\bar{\varepsilon} = \frac{1}{n} \sum_{i=1}^n (1 - \hat{p}_{y_i^*}(x_i))$  be the mean complement of the model’s voted-class confidence on the calibration set. Define the pseudo-target distribution

$$\tilde{\pi}_i^{\text{LS}} = (1 - \bar{\varepsilon}) e_{y_i^*} + \frac{\bar{\varepsilon}}{K} \mathbf{1}, \quad (4)$$

and minimize  $\mathcal{L}_{\text{SLTS}}$  (Eq. 3) with these targets. The global smoothing weight  $\bar{\varepsilon}$  is a data-driven estimate of average annotator disagreement: if the model is 80% confident on the voted class, the remaining 20% is spread uniformly across all  $K$  classes. The result is  $T_{\text{LS}}^* > T_{\text{TS}}^*$  whenever  $\bar{\varepsilon} > 0$ , moving the calibrator in the correct direction even without annotator data.

**Theoretical interpretation.** The smoothing weight  $\bar{\varepsilon}$  satisfies a moment-matching fixed point: the expected pseudo-label mass on the voted class equals the average model confidence,  $\mathbb{E}[\tilde{\pi}_{y^*}^{\text{LS}}] = \mathbb{E}[\hat{p}_{y^*}]$ . Algorithmically, LS-TS can be viewed as a single E-step of an EM algorithm in which the latent annotator distribution is estimated from the current (pre-calibration) model, followed by one M-step that optimises  $T$  given those pseudo-labels. This connection to self-distillation [Hinton et al., 2015] explains the self-referential nature of the approach: the uncalibrated model both supplies the pseudo-targets and is then calibrated against them.

## 7 Experiments

We evaluate on four benchmarks with multi-annotator data, each with two backbones: CIFAR-10H (ResNet-50, ViT-B/16), ChaosNLI (RoBERTa-Large, DeBERTa-v3), ISIC 2019 (EfficientNet-B4, ViT-S/16), and DermaMNIST (ResNet-18, ViT-S/16).

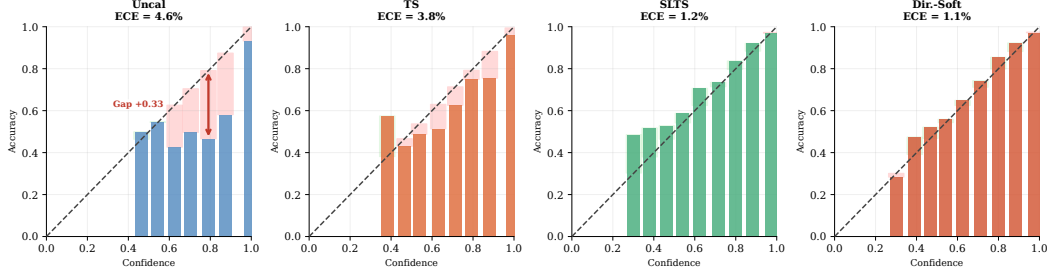


Figure 3: **Reliability diagrams for four representative methods — CIFAR-10H ResNet-50 ( $ECE_{\text{true}}$ )**. Red shading indicates overconfidence; green indicates underconfidence. Uncal and TS remain overconfident; SLTS substantially corrects this with a single temperature parameter; Dirichlet-Soft reduces the residual gap further. Reliability diagrams for all remaining methods are provided in Appendix H.

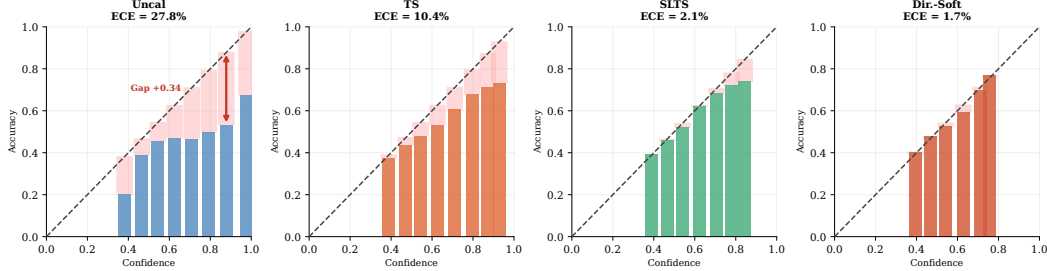


Figure 4: **Reliability diagrams for four representative methods — ChaosNLI RoBERTa-Large ( $ECE_{\text{true}}$ )**. Same method selection as Figure 3. Red shading indicates overconfidence; green indicates underconfidence. NLI models are severely overconfident before calibration; ambiguity-aware methods substantially correct this.

182 **Baselines.** We compare against (i) **Uncalibrated**: raw softmax; (ii) **TS**: Temperature Scaling on  
 183 voted labels [Guo et al., 2017]; (iii) **Platt (PS)**: per-class weight and bias on logits, calibrated against  
 184 voted labels [Platt, 1999, Kull et al., 2019]; (iv) **Dirichlet-Hard**: Dirichlet calibration [Kull et al.,  
 185 2019] with the full  $K \times K$  weight matrix and bias, calibrated against voted labels. Dirichlet-Hard  
 186 uses the *same architecture* as our Dirichlet-Soft but with voted-label targets; the contrast isolates the  
 187 effect of the calibration target from model capacity.

188 **Metrics.** All metrics are evaluated against true labels  $\tilde{y} \sim \pi(\cdot | x)$ :  $ECE_{\text{true}}$  (averaged over 100  
 189 draws;  $B = 15$  equal-width bins), adaptive ECE ( $aECE_t$ ; equal-mass bins, same draws), classwise  
 190 ECE ( $cwECE_t$ ; per-class ECE averaged over  $K$  classes [Kull et al., 2019]), Brier score ( $Br_t$ ), and  
 191 NLL ( $NLL_t$ ). We emphasize  $ECE_{\text{true}}$  because it mirrors the evaluation protocol a practitioner faces  
 192 when only one label is observed at test time; the Brier score and NLL directly measure distributional  
 193 alignment with  $\hat{\pi}$  and serve as complementary distributional metrics. We abbreviate  $ECE_{\text{true}}$ ,  $aECE$ ,  
 194  $cwECE$ ,  $Br$ ,  $NLL$  throughout.

## 195 7.1 CIFAR-10H: Image Classification with Human Label Noise

196 **Dataset and models.** CIFAR-10H [Peterson et al., 2019] provides  $\approx 51$  human annotations per  
 197 image for the 10 000-image CIFAR-10 test set, directly exposing perceptual ambiguity in low-  
 198 resolution natural images through repeated human labeling. The test set is split into calibration  
 199 ( $n = 5,000$ ) and evaluation (5,000) by stratified sampling (seed 42). We evaluate two architectures:  
 200 **ResNet-50** [He et al., 2016] pretrained on ImageNet-1k, fine-tuned for 30 epochs (AdamW, cosine  
 201 annealing), achieving 97.3% top-1 accuracy. **ViT-B/16** [Dosovitskiy et al., 2021] pretrained on  
 202 ImageNet-21k, fine-tuned with the same protocol, achieving 98.1% top-1 accuracy.

Table 1: **CIFAR-10H and ChaosNLI results.** All metrics evaluated against true (sampled) labels. Best per architecture in **bold**.

| Method                                                           | ResNet-50     |             |             |             |              |              | ViT-B/16   |             |             |             |              |              |
|------------------------------------------------------------------|---------------|-------------|-------------|-------------|--------------|--------------|------------|-------------|-------------|-------------|--------------|--------------|
|                                                                  | $T$           | ECE↓        | aECE↓       | cwECE↓      | Br↓          | NLL↓         | $T$        | ECE↓        | aECE↓       | cwECE↓      | Br↓          | NLL↓         |
| <b>CIFAR-10H</b> ( $\approx 51$ annotations per image, $K=10$ )  |               |             |             |             |              |              |            |             |             |             |              |              |
| <i>Voted-label baselines</i>                                     |               |             |             |             |              |              |            |             |             |             |              |              |
| Uncalibrated                                                     | —             | 4.97        | 5.71        | 1.09        | 0.120        | 0.692        | —          | 4.99        | 5.36        | 1.06        | 0.111        | 0.678        |
| TS                                                               | 2.03          | 4.29        | 4.25        | 0.91        | 0.112        | 0.363        | 2.04       | 4.48        | 4.40        | 0.93        | 0.106        | 0.346        |
| Platt                                                            | —             | 4.29        | 4.23        | 0.91        | 0.112        | 0.372        | —          | 4.54        | 4.38        | 0.93        | 0.105        | 0.363        |
| Dirichlet-Hard                                                   | —             | 4.46        | 4.44        | 0.95        | 0.114        | 0.395        | —          | 4.70        | 4.62        | 0.97        | 0.107        | 0.394        |
| <i>Ambiguity-aware (ours)</i>                                    |               |             |             |             |              |              |            |             |             |             |              |              |
| SLTS                                                             | 3.18          | 1.51        | 1.39        | 0.45        | 0.110        | 0.293        | 3.07       | 0.85        | 0.56        | 0.39        | 0.102        | 0.278        |
| SoftPlatt                                                        | —             | 1.52        | 1.31        | <b>0.35</b> | 0.110        | 0.288        | —          | 0.88        | 0.47        | <b>0.24</b> | 0.101        | 0.272        |
| VS                                                               | —             | 1.35        | 1.26        | 0.39        | <b>0.109</b> | 0.289        | —          | 0.92        | 0.50        | 0.32        | 0.101        | 0.274        |
| IR-Soft                                                          | —             | <b>0.72</b> | <b>0.83</b> | 0.45        | 0.115        | 0.340        | —          | 0.91        | 0.61        | 0.43        | 0.105        | 0.321        |
| Dirichlet-Soft                                                   | —             | 1.25        | 1.06        | <b>0.35</b> | <b>0.109</b> | <b>0.271</b> | —          | <b>0.72</b> | <b>0.41</b> | 0.25        | <b>0.100</b> | <b>0.255</b> |
| <i>Annotation-free (ours)</i>                                    |               |             |             |             |              |              |            |             |             |             |              |              |
| LS-TS                                                            | 3.08          | 1.57        | 1.31        | 0.46        | 0.109        | 0.293        | 2.76       | 2.37        | 2.21        | 0.53        | 0.103        | 0.285        |
| Method                                                           | RoBERTa-Large |             |             |             |              |              | DeBERTa-v3 |             |             |             |              |              |
|                                                                  | $T$           | ECE↓        | aECE↓       | cwECE↓      | Br↓          | NLL↓         | $T$        | ECE↓        | aECE↓       | cwECE↓      | Br↓          | NLL↓         |
| <b>ChaosNLI</b> (SNLI+MNLI, 100 annotations per example, $K=3$ ) |               |             |             |             |              |              |            |             |             |             |              |              |
| <i>Voted-label baselines</i>                                     |               |             |             |             |              |              |            |             |             |             |              |              |
| Uncalibrated                                                     | —             | 27.79       | 27.77       | 18.61       | 0.650        | 1.384        | —          | 35.97       | 35.92       | 24.02       | 0.743        | 2.148        |
| TS                                                               | 2.42          | 10.55       | 10.38       | 7.30        | 0.537        | 0.886        | 3.81       | 11.63       | 11.57       | 8.48        | 0.549        | 0.900        |
| Platt                                                            | —             | 11.16       | 10.92       | 7.28        | 0.530        | 0.875        | —          | 12.43       | 12.36       | 8.43        | 0.539        | 0.896        |
| Dirichlet-Hard                                                   | —             | 11.55       | 11.40       | 7.64        | 0.535        | 0.889        | —          | 12.69       | 12.63       | 8.84        | 0.539        | 0.895        |
| <i>Ambiguity-aware (ours)</i>                                    |               |             |             |             |              |              |            |             |             |             |              |              |
| SLTS                                                             | 3.41          | 3.22        | 2.59        | 4.62        | 0.520        | 0.862        | 5.28       | 3.45        | 3.36        | 6.53        | 0.529        | 0.877        |
| SoftPlatt                                                        | —             | 2.66        | 2.32        | 2.43        | <b>0.509</b> | <b>0.839</b> | —          | 3.17        | 2.92        | 3.18        | 0.511        | 0.841        |
| VS                                                               | —             | 3.46        | 3.07        | 5.13        | 0.518        | 0.857        | —          | 3.91        | 3.35        | 6.46        | 0.525        | 0.869        |
| IR-Soft                                                          | —             | 2.65        | 2.59        | 6.24        | 0.549        | 0.934        | —          | <b>2.15</b> | 3.96        | 7.20        | 0.554        | 0.942        |
| Dirichlet-Soft                                                   | —             | <b>2.57</b> | <b>1.71</b> | <b>2.03</b> | 0.510        | <b>0.839</b> | —          | 3.19        | <b>2.71</b> | <b>2.95</b> | <b>0.509</b> | <b>0.836</b> |
| <i>Annotation-free (ours)</i>                                    |               |             |             |             |              |              |            |             |             |             |              |              |
| LS-TS                                                            | 4.40          | 4.16        | 3.72        | 5.61        | 0.522        | 0.873        | 6.15       | 2.65        | 2.99        | 6.32        | 0.529        | 0.881        |

**Results.** Both CIFAR-10H backbones show the same pattern: hard-label baselines still leave 4–5% ECE<sub>true</sub>, while ambiguity-aware methods reduce it to below 2%. Dirichlet-Soft achieves the best Brier score and NLL on both architectures (0.109/0.271 on ResNet-50; 0.100/0.255 on ViT-B/16), confirming that expressive distribution-aware calibrators provide the best distributional fit. LS-TS, using only voted labels, already reduces ECE from 4.29% to 1.57% (ResNet-50) and from 4.48% to 2.37% (ViT-B/16), competitive with fully ambiguity-aware methods.

## 7.2 ChaosNLI: Natural Language Inference

We next evaluate on **ChaosNLI** [Nie et al., 2020]: 100 human annotations per example for the combined SNLI+MNLI subset ( $n = 3,113$ ), where disagreement reflects genuine semantic ambiguity rather than annotation noise. The data are split 50/50 stratified by voted label. We evaluate **RoBERTa-Large** [Liu et al., 2019] and **DeBERTa-v3-base** [He et al., 2023], both fine-tuned on MNLI.

**Results.** The same pattern holds in NLI: hard-label baselines leave 10–12% ECE, while Dirichlet-Soft achieves the best overall distributional fit (Brier 0.510/0.509; NLL 0.839/0.836 on RoBERTa-L/DeBERTa-v3). LS-TS reduces ECE from 10.55% to 4.16% (RoBERTa-L) and from 11.63% to 2.65% (DeBERTa-v3) using only voted labels, outperforming all hard-label baselines by a wide margin.

## 7.3 ISIC 2019: Skin Disease Diagnosis

We evaluate on **ISIC 2019** [Codella et al., 2019, Combalia et al., 2019] (8 skin conditions,  $\approx 25,000$  images), a clinically realistic differential-diagnosis task in which several lesion categories have

Table 2: **ISIC 2019 and DermaMNIST results.** All metrics evaluated against true (sampled) labels. Best per architecture in **bold**.

| Method                                                                      | EfficientNet-B4 |             |             |             |              |              | ViT-S/16 |             |             |             |              |              |
|-----------------------------------------------------------------------------|-----------------|-------------|-------------|-------------|--------------|--------------|----------|-------------|-------------|-------------|--------------|--------------|
|                                                                             | $T$             | ECE↓        | aECE↓       | cwECE↓      | Br↓          | NLL↓         | $T$      | ECE↓        | aECE↓       | cwECE↓      | Br↓          | NLL↓         |
| <b>ISIC 2019</b> ( $m=9$ synthetic annotators, mean diagonal agreement 75%) |                 |             |             |             |              |              |          |             |             |             |              |              |
| <i>Voted-label baselines</i>                                                |                 |             |             |             |              |              |          |             |             |             |              |              |
| Uncalibrated                                                                | —               | 25.76       | 25.84       | 6.67        | 0.600        | 2.710        | —        | 23.31       | 23.33       | 6.55        | 0.651        | 1.978        |
| TS                                                                          | 1.79            | 18.54       | 18.49       | 5.02        | 0.560        | 1.653        | 1.42     | 16.59       | 16.60       | 5.23        | 0.617        | 1.563        |
| Platt                                                                       | —               | 19.89       | 19.81       | 5.24        | 0.562        | 1.659        | —        | 17.19       | 17.17       | 4.74        | 0.597        | 1.583        |
| Dirichlet-Hard                                                              | —               | 20.36       | 20.32       | 5.33        | 0.563        | 1.674        | —        | 17.76       | 17.72       | 4.86        | 0.600        | 1.597        |
| <i>Ambiguity-aware (ours)</i>                                               |                 |             |             |             |              |              |          |             |             |             |              |              |
| SLTS                                                                        | 4.46            | 9.71        | 9.74        | 2.92        | 0.528        | 1.146        | 3.15     | 6.96        | 6.68        | 3.29        | 0.586        | 1.233        |
| SoftPlatt                                                                   | —               | 9.25        | 9.31        | 2.23        | 0.523        | 1.108        | —        | 8.05        | 7.90        | 2.05        | 0.565        | 1.190        |
| VS                                                                          | —               | 9.03        | 9.11        | 2.42        | 0.524        | 1.121        | —        | 8.50        | 8.59        | 2.24        | 0.567        | 1.198        |
| IR-Soft                                                                     | —               | <b>1.75</b> | <b>2.43</b> | 4.28        | 0.538        | 1.284        | —        | <b>1.73</b> | <b>1.59</b> | 5.78        | 0.621        | 1.466        |
| Dirichlet-Soft                                                              | —               | 7.50        | 7.48        | <b>2.00</b> | <b>0.518</b> | <b>1.084</b> | —        | 6.85        | 6.80        | <b>1.77</b> | <b>0.563</b> | <b>1.175</b> |
| <i>Annotation-free (ours)</i>                                               |                 |             |             |             |              |              |          |             |             |             |              |              |
| LS-TS                                                                       | 4.30            | 9.34        | 9.46        | 2.98        | 0.528        | 1.149        | 3.98     | 15.49       | 15.47       | 4.52        | 0.619        | 1.303        |
| Method                                                                      | ResNet-18       |             |             |             |              |              | ViT-S/16 |             |             |             |              |              |
|                                                                             | $T$             | ECE↓        | aECE↓       | cwECE↓      | Br↓          | NLL↓         | $T$      | ECE↓        | aECE↓       | cwECE↓      | Br↓          | NLL↓         |
| <b>DermaMNIST</b> ( $m=5$ synthetic annotators, overall agreement 64.7%)    |                 |             |             |             |              |              |          |             |             |             |              |              |
| <i>Voted-label baselines</i>                                                |                 |             |             |             |              |              |          |             |             |             |              |              |
| Uncalibrated                                                                | —               | 34.35       | 34.26       | 10.15       | 0.767        | 2.689        | —        | 37.12       | 37.10       | 10.82       | 0.786        | 3.766        |
| TS                                                                          | 1.86            | 23.05       | 22.82       | 6.97        | 0.686        | 1.654        | 2.60     | 24.36       | 24.26       | 7.29        | 0.683        | 1.664        |
| Platt                                                                       | —               | 23.64       | 23.56       | 7.08        | 0.682        | 1.667        | —        | 23.82       | 23.59       | 7.10        | 0.675        | 1.681        |
| Dirichlet-Hard                                                              | —               | 24.58       | 24.46       | 7.33        | 0.690        | 1.771        | —        | 24.84       | 24.68       | 7.49        | 0.682        | 1.897        |
| <i>Ambiguity-aware (ours)</i>                                               |                 |             |             |             |              |              |          |             |             |             |              |              |
| SLTS                                                                        | 3.51            | 4.61        | 4.44        | 3.56        | 0.622        | 1.376        | 5.11     | 3.58        | 3.06        | 3.37        | 0.610        | 1.350        |
| SoftPlatt                                                                   | —               | 4.19        | 3.55        | 1.61        | 0.612        | 1.312        | —        | 3.56        | 2.93        | 1.29        | <b>0.601</b> | 1.289        |
| VS                                                                          | —               | 4.45        | 3.83        | 3.23        | 0.621        | 1.363        | —        | 4.03        | 3.71        | 2.84        | 0.611        | 1.331        |
| IR-Soft                                                                     | —               | <b>2.15</b> | <b>2.34</b> | 4.55        | 0.634        | 1.440        | —        | 3.79        | 3.73        | 4.59        | 0.628        | 1.431        |
| Dirichlet-Soft                                                              | —               | 3.55        | 3.36        | <b>1.22</b> | <b>0.610</b> | <b>1.305</b> | —        | <b>3.32</b> | <b>2.87</b> | <b>1.14</b> | <b>0.601</b> | <b>1.286</b> |
| <i>Annotation-free (ours)</i>                                               |                 |             |             |             |              |              |          |             |             |             |              |              |
| LS-TS                                                                       | 3.03            | 5.05        | 4.82        | 3.79        | 0.630        | 1.394        | 4.19     | 7.36        | 7.06        | 3.68        | 0.615        | 1.362        |

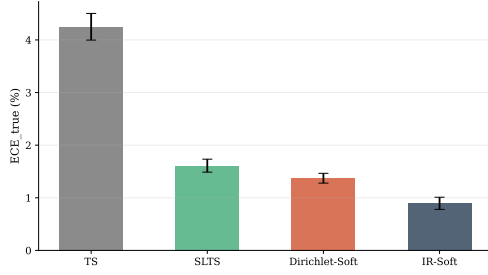
overlapping visual morphology. We use  $m = 9$  synthetic dermatologist annotations per image, generated from a clinically calibrated  $8 \times 8$  confusion matrix (Appendix I) matching Liu et al. [2020]: mean diagonal agreement 75%. We evaluate **EfficientNet-B4** [Tan and Le, 2019] and **ViT-S/16** [Dosovitskiy et al., 2021] on a stratified 70/15/15 split.

**ISIC 2019 results.** The medical setting shows the same failure mode for standard calibrators and the same benefit from ambiguity-aware targets. Dirichlet-Soft again achieves the best Brier and NLL on both backbones (0.518/1.084 on ENet-B4; 0.563/1.175 on ViT-S/16), while IR-Soft achieves the lowest ECE. LS-TS halves the ECE of TS on ENet-B4 (18.54%→9.34%) using only voted labels; the improvement is smaller on ViT-S/16 (16.59%→15.49%), highlighting that model confidence is not always a reliable proxy for annotation ambiguity.

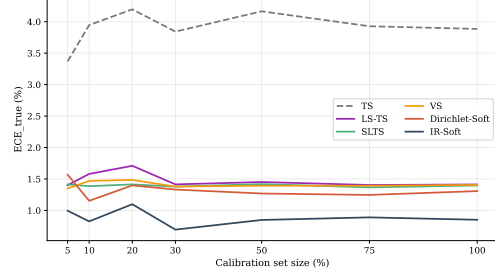
#### 7.4 DermaMNIST: Supplementary Skin Disease Experiment

We additionally evaluate on **DermaMNIST** [Yang et al., 2023] (7-class, derived from HAM10000 [Tschandl et al., 2018];  $m = 5$  synthetic annotators; overall agreement  $\approx 64.7\%$ ) with two backbones: ResNet-18 and ViT-S/16. Results in Table 2 show that TS leaves  $ECE = 23.1\%$  for ResNet-18 while ambiguity-aware methods reduce it substantially: SLTS achieves 4.6% (80% reduction), IR-Soft achieves 2.2% (91% reduction), and Dirichlet-Soft achieves 3.6% while attaining the best cwECE (1.2%), Brier (0.610), and NLL (1.305). ViT-S/16 shows a similar pattern (SLTS: 3.6%; Dirichlet-Soft achieves the best ECE, Brier, and NLL: 3.3%/0.601/1.286). Both backbones show  $T_{\text{SLTS}} > T_{\text{TS}} > 1$ : SLTS requires substantially higher temperature to match the more diffuse annotation distribution.





(a) Multi-seed stability. Mean  $ECE_{true}$  ( $\pm$ std) over 5 seeds; ranking is stable across splits.



(b) Calibration-set-size ablation.  $ECE_{true}$  vs. fraction of calibration data; ambiguity-aware methods converge at 5–10%.

Figure 5: **Robustness of ambiguity-aware calibration — CIFAR-10H ResNet-50.** Full results for all architectures in Appendices G and E; annotation-count ablation in Appendix F.

## 7.5 Reliability Diagrams

Figure 3 shows reliability diagrams for four representative methods on CIFAR-10H (ResNet-50). The diagrams are evaluated against the true label distribution ( $ECE_{true}$ ). Uncal and TS are systematically overconfident: bars fall well below the diagonal at high confidence, leaving a large red gap. SLTS corrects the overconfidence with a scalar temperature adjustment, aligning most bins close to the diagonal. Dirichlet-Soft achieves near-perfect calibration by combining the expressive  $K \times K$  affine transform with the correct distributional target. Figure 4 shows corresponding diagrams for ChaosNLI (RoBERTa-Large); full  $3 \times 5$  panels covering all 13 methods and all remaining architectures are provided in Appendix H.

## 8 Discussion

**Sacrificing voted-label calibration.** A natural question is whether ambiguity-aware methods degrade  $ECE_{voted}$ . This is expected behaviour: voted-label calibration is epistemically unreliable when annotators genuinely disagree, as the voted label overstates the majority-class probability. Practitioners who must satisfy a hard  $ECE_{voted}$  constraint can apply TS on top of Dirichlet-Soft or SLTS outputs.

**Practical guidelines.** Use **Dirichlet-Soft** as the default when annotator distributions are available — it achieves the best Brier score and NLL across all benchmarks while maintaining competitive ECE. Use **SLTS** when a single-parameter method is preferred for simplicity; **IR-Soft** when the lowest ECE is the sole criterion; and **LS-TS** when the calibration set contains only one-hot voted labels.

**Robustness.** Figure 5a confirms that the ranking of methods is stable across random seeds ( $\text{std} \leq 0.54$  pp); full results for all architectures are in Appendix G. Figure 5b shows that ambiguity-aware methods converge with as few as 5–10% of the calibration set; full results for all architectures are in Appendix E.

**Limitations.** Full ambiguity-aware calibration (including Dirichlet-Soft) requires multi-annotator labels at calibration time. When only voted labels are available, LS-TS remains useful but does not fully match Dirichlet-Soft, particularly on datasets where model confidence diverges from annotation ambiguity (e.g., ISIC 2019 ViT-S/16). ISIC 2019 and DermaMNIST rely on synthetic annotators derived from clinician-reported agreement, so real multi-reader medical calibration sets would provide stronger validation.

## 9 Conclusion

We formalized the problem of confidence calibration under ambiguous ground truth and showed that standard post-hoc calibrators fitted on voted labels are systematically miscalibrated against the

underlying ambiguous label distribution. Our flagship ambiguity-aware method, **Dirichlet-Soft**, requires no retraining and achieves the best Brier score and NLL across all four benchmarks and eight architectures spanning vision and language, reducing true-label ECE by 59–86% relative to Temperature Scaling. The key insight is that the calibration target — not method capacity — is the limiting factor: Dirichlet calibration with voted-label targets (Dirichlet-Hard) consistently *underperforms* simple TS, while the same architecture with soft-label targets (Dirichlet-Soft) achieves the best overall results. Crucially, when the calibration set contains only one-hot voted labels, our annotation-free method **LS-TS** still improves true-label calibration, reducing ECE by 7–78% using only the model’s own confidence as a proxy for annotator disagreement. We recommend Dirichlet-Soft as the default wherever multi-annotator data is available, and LS-TS as a practical post-hoc alternative when only voted labels are available.

**Acknowledgements.** We thank the anonymous reviewers for their constructive feedback.

## References

- J. Baan, W. Aziz, B. Plank, and R. Fernández. Stop measuring calibration when humans disagree. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1892–1915, 2022.
- Y. Bai, S. Mei, H. Wang, and C. Xiong. Don’t just blame over-parametrization for over-confidence: Theoretical analysis of calibration in binary classification. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, pages 566–576. PMLR, 2021.
- N. Codella, V. Rotemberg, P. Tschandl, M. E. Celebi, S. Dusza, D. Gutman, B. Helba, A. Kalloo, K. Liopyris, M. Marchetti, H. Kittler, and A. Halpern. Skin lesion analysis toward melanoma detection: ISIC 2018 challenge. In *arXiv preprint arXiv:1902.03368*, 2019.
- M. Combalia, N. C. F. Codella, V. Rotemberg, B. Helba, V. Vilaplana, O. Reiter, C. Carrera, A. Barreiro, A. C. Halpern, S. Puig, and J. Malvehy. BCN20000: Dermoscopic lesions in the wild. *arXiv preprint arXiv:1908.02288*, 2019.
- A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. *International Conference on Learning Representations (ICLR)*, 2021.
- M. L. Gordon, M. S. Lam, J. S. Park, K. Patel, J. Hancock, T. Hashimoto, and M. S. Bernstein. Jury learning: Integrating dissenting voices into machine learning models. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 2022.
- C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, pages 1321–1330. PMLR, 2017.
- H. A. Haenssle, C. Fink, R. Schneiderbauer, F. Toberer, T. Buhl, A. Blum, A. Kalloo, A. B. H. Hassen, L. Thomas, A. Enk, W. Stolz, and Reader Study Level-I and Level-II Investigators. Man against machine: Diagnostic performance of a deep learning convolutional neural network for dermoscopic melanoma recognition in comparison to 58 dermatologists. *Annals of Oncology*, 29(8):1836–1842, 2018.
- K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.
- P. He, J. Gao, and W. Chen. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. In *International Conference on Learning Representations (ICLR)*, 2023.
- G. Hinton, O. Vinyals, and J. Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- U. Khurana, E. Nalisnick, A. Fokkens, and S. Swayamdipta. Crowd-calibrator: Can annotator disagreement inform calibration in subjective tasks? In *Proceedings of the Conference on Language Modeling (COLM)*, 2024.
- B. Kompa, J. Snoek, and A. L. Beam. Second opinion needed: Communicating uncertainty in medical machine learning. *NPJ Digital Medicine*, 4(1):4, 2021.
- M. Kull, M. P. Nieto, M. Kängsepp, T. Silva Filho, H. Song, and P. Flach. Beyond temperature scaling: Obtaining well-calibrated multi-class probabilities with Dirichlet calibration. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.
- Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov. RoBERTa: A robustly optimized BERT pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.

332 Y. Liu, A. Jain, C. Eng, D. H. Way, K. Lee, P. Bui, K. Kanada, G. de Oliveira Marinho, J. Gallegos,  
333 S. Gabriele, V. Gupta, N. Singh, V. Natarajan, R. Hofmann-Wellenhof, G. S. Corrado, L. H. Peng,  
334 D. R. Webster, D. Ai, S. Huang, Y. Liu, R. C. Dunn, and D. Coz. A deep learning system for  
335 differential diagnosis of skin diseases. *Nature Medicine*, 26:900–908, 2020.

336 M. Minderer, J. Djolonga, R. Romijnders, F. A. Hubis, X. Zhai, N. Houlsby, D. Tran, and M. Lucic.  
337 Revisiting the calibration of modern neural networks. In *Advances in Neural Information  
338 Processing Systems (NeurIPS)*, volume 34, 2021.

339 R. Müller, S. Kornblith, and G. Hinton. When does label smoothing help? In *Advances in Neural  
340 Information Processing Systems (NeurIPS)*, volume 32, 2019.

341 Y. Nie, X. Zhou, and M. Bansal. What can we learn from collective human opinions on natural  
342 language inference data? In *Proceedings of the 2020 Conference on Empirical Methods in Natural  
343 Language Processing (EMNLP)*, pages 9131–9143, 2020.

344 S. Park, O. Bastani, J. Weimer, and I. Lee. Calibrated prediction with covariate shift via unsupervised  
345 domain adaptation. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*,  
346 pages 3219–3229. PMLR, 2020.

347 J. C. Peterson, R. M. Battleday, T. L. Griffiths, and O. Russakovsky. Human uncertainty makes  
348 classification more robust. In *Proceedings of the IEEE/CVF International Conference on Computer  
349 Vision (ICCV)*, pages 9617–9626, 2019.

350 B. Plank. The problem with human label variation: On ground truth in data, modeling and evaluation.  
351 *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing  
352 (EMNLP)*, 2022.

353 J. C. Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood  
354 methods. In *Advances in Large Margin Classifiers*, pages 61–74. MIT Press, 1999.

355 F. Rodrigues and F. Pereira. Deep learning from crowds. In *Proceedings of the 32nd AAAI Conference  
356 on Artificial Intelligence*, 2018.

357 A. Singh, A. Tiwari, H. Hasanbeig, and P. Gupta. Soft-label training preserves epistemic uncertainty.  
358 *arXiv preprint arXiv:2511.14117*, 2025.

359 D. Stutz, A. G. Roy, T. Matejovicova, P. Strachan, A. T. Cemgil, and A. Doucet. Conformal  
360 prediction under ambiguous ground truth. *Transactions on Machine Learning Research (TMLR)*,  
361 2023. arXiv:2307.09302.

362 C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna. Rethinking the inception architecture  
363 for computer vision. In *Proceedings of the IEEE Conference on Computer Vision and Pattern  
364 Recognition (CVPR)*, pages 2818–2826, 2016.

365 M. Tan and Q. V. Le. EfficientNet: Rethinking model scaling for convolutional neural networks. In  
366 *Proceedings of the 36th International Conference on Machine Learning (ICML)*, pages 6105–6114,  
367 2019.

368 S. Thulasidasan, G. Chennupati, J. A. Bilmes, T. Bhattacharya, and S. Michalak. On mixup training:  
369 Improved calibration and predictive uncertainty for deep neural networks. In *Advances in Neural  
370 Information Processing Systems (NeurIPS)*, volume 32, 2019.

371 P. Tschandl, C. Rosendahl, and H. Kittler. The HAM10000 dataset, a large collection of multi-source  
372 dermatoscopic images of common pigmented skin lesions. *Scientific Data*, 5:180161, 2018.

373 A. N. Uma, T. Fornaciari, D. Hovy, E. Pavlick, M. Poesio, and B. Plank. Learning from disagreement:  
374 A survey. *Journal of Artificial Intelligence Research*, 72:1385–1470, 2021.

375 X. Wang, M. Long, J. Wang, and M. Jordan. Transferable calibration with lower bias and variance in  
376 domain adaptation. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33,  
377 2020.

- 378 J. Yang, R. Shi, D. Wei, Z. Liu, L. Zhao, B. Ke, H. Pfister, and B. Ni. MedMNIST v2: A large-scale  
379 lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*, 10(1):41,  
380 2023.
- 381 B. Zadrozny and C. Elkan. Transforming classifier scores into accurate multiclass probability  
382 estimates. In *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge*  
383 *Discovery and Data Mining*, pages 694–699, 2002.

## A Dataset Background and Sources of Label Ambiguity

All four benchmarks in this paper are appropriate for studying calibration under ambiguous ground truth, but they exhibit ambiguity in slightly different ways.

**CIFAR-10H.** Peterson et al. [2019] introduced CIFAR-10H by collecting a full distribution of human labels for each image in the CIFAR-10 test set, explicitly to capture *human perceptual uncertainty*. The ambiguous examples are typically low-resolution images whose visual evidence supports multiple plausible classes for human observers. In this benchmark, the label distribution is therefore directly observed from repeated human annotation, making it a clean testbed for ambiguity-aware calibration.

**ChaosNLI.** Nie et al. [2020] introduced ChaosNLI to study *collective human opinions* on natural language inference. Each example receives 100 human labels, and the paper reports that substantial disagreement is common rather than exceptional. This ambiguity is semantic rather than perceptual: differences in pragmatic assumptions, underspecification, and sentence interpretation lead reasonable annotators to assign different NLI labels to the same premise–hypothesis pair.

**ISIC 2019.** ISIC 2019 is a multiclass dermoscopic lesion diagnosis benchmark assembled from several clinical sources, including BCN20000 [Codella et al., 2019, Combalia et al., 2019]. The task itself is ambiguity-prone because several lesion categories share overlapping visual morphology, and the challenge was designed to reflect the more realistic task of differential diagnosis rather than a simpler benign-versus-malignant decision. Because the public challenge release does not provide per-image reader distributions, we model ambiguity using synthetic annotators calibrated to dermatologist confusion patterns and reader agreement reported in Liu et al. [2020]. This preserves the core property relevant to our study: in dermatology, a single image can support multiple clinically plausible labels.

**DermaMNIST.** DermaMNIST [Yang et al., 2023] is a standardized  $28 \times 28$  version of the dermatology benchmark derived from HAM10000 [Tschandl et al., 2018]. HAM10000 consists of dermoscopic images of common pigmented lesions collected in clinical practice, where diagnosis already involves fine-grained distinctions among visually similar categories such as melanoma, nevi, and benign keratoses. DermaMNIST inherits this medical ambiguity from HAM10000 and, because it is aggressively downsampled for lightweight benchmarking, removes additional visual detail that can further increase label uncertainty. As with ISIC 2019, we therefore evaluate calibration against an ambiguity-aware label distribution rather than treating the majority diagnosis as uniquely correct.

## 415 B Proof of Proposition 1

416 Let  $\mathcal{D}_{\text{amb}} = \{(x_i, y_i^*) : y_i^* = c, x_i \in \mathcal{C}\}$  be the subset of calibration examples from the ambiguous  
 417 cluster  $\mathcal{C}$ , where the voted label is always class  $c$ . Denote  $p_i = \text{softmax}(z_i)_c$ .

418 The TS loss on  $\mathcal{D}_{\text{amb}}$  is  $\mathcal{L}_{\text{TS}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \log \text{softmax}(z_i/T)_c$ . As  $T \rightarrow 0^+$ ,  
 419  $\text{softmax}(z_i/T)_c \rightarrow 1$  (assuming  $z_{ic} > z_{ik}$  for all  $k \neq c$ ), so  $\mathcal{L}_{\text{TS}} \rightarrow 0$ . Hence  $\partial \mathcal{L}_{\text{TS}} / \partial T > 0$   
 420 for all  $T > 0$ : decreasing  $T$  always decreases the voted-label loss. The optimal  $T^*$  balances con-  
 421 tributions from ambiguous and unambiguous examples, but  $T^* \leq T_{\text{no-amb}}^*$  (the optimal temperature  
 422 without the ambiguous cluster). In particular,  $T^* < 1$  whenever the model is already reasonably  
 423 accurate on the ambiguous cluster ( $p_i > q$  on average).

424 The SLTS loss on  $\mathcal{D}_{\text{amb}}$  is  $\mathcal{L}_{\text{SLTS}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \sum_k \hat{\pi}_k(x_i) \log \text{softmax}(z_i/T)_k$ , with  
 425  $\hat{\pi}_c(x_i) = q < 1$ . The minimum-loss  $T^*$  satisfies  $\text{softmax}(z_i/T^*)_c \approx q$ , requiring  $T^* > 1$  whenever  
 426  $p_i > q$ .  $\square$

## C Proof of Proposition 2

We formalize the argument given in the proof sketch in the main text.

**Setup.** Let  $p = \hat{p}_{\hat{c}}(x)$  denote the model’s predicted confidence for the top class  $\hat{c}$  and let  $q = \pi_{\hat{c}}(x) \in (0, 1]$  denote the true annotator probability for that class. We assume the model is trained against voted labels, so  $p$  is approximately constant across examples with the same voted label regardless of their annotation entropy  $H(x)$ .

**Voted-label calibration error contribution.** For example  $x_i$  in confidence bin  $B_b$  (where  $\bar{p}(B_b) \approx p$ ), the per-example voted-label calibration error is  $|p - \mathbf{1}[\hat{c} = y^*]|$ . Since  $\hat{c} = y^*$  for correctly-classified ambiguous examples (the model predicts the majority class), this contribution is approximately  $|p - 1|$  for examples where the model makes the right hard prediction, independent of  $H(x)$ .

**True-label calibration error contribution.** The per-example true-label calibration error is  $|p - \pi_{\hat{c}}(x)| = |p - q|$ .

**Excess true-label error relative to voted-label evaluation.** The per-example excess contribution is:

$$\delta(x) = |p - q| - |p - \mathbf{1}[\hat{c} = y^*]|. \quad (5)$$

When the model is correctly calibrated to voted labels ( $p \approx \mathbf{1}[\hat{c} = y^*]$  in expectation), the voted contribution is near zero:  $|p - 1| \approx 0$  for high-accuracy bins and  $|p - 0| = p$  for error bins. For unambiguous examples ( $H(x) = 0$ ),  $q = \pi_{y^*} = 1$ , so  $\delta(x) = |p - 1| - |p - 1| = 0$ .

**Monotonicity.** As  $H(x)$  increases,  $q = \pi_{y^*}(x)$  decreases below 1 (more probability mass shifts to non-majority classes). Specifically, for a  $K$ -class problem with uniform annotator distribution at maximum entropy,  $q = 1/K$ . The true-label contribution  $|p - q|$  increases strictly as  $q$  decreases from 1 (since  $p > q$  for a well-trained model predicting the majority class with confidence exceeding the majority probability). The voted-label contribution  $|p - \mathbf{1}[\hat{c} = y^*]|$  is unaffected by  $H(x)$ . Therefore  $\delta(x) = |p - q| - |p - \mathbf{1}[\hat{c} = y^*]|$  is non-decreasing in  $H(x)$ , and strictly increasing whenever  $p > q > 0$ .

**Aggregate monotonicity.** The aggregate excess true-label error relative to voted-label evaluation equals  $\sum_b \frac{|B_b|}{n} \sum_{i \in B_b} \delta(x_i) / |B_b|$ . Since  $\delta(x_i)$  is non-decreasing in  $H(x_i)$  and bins contain a mixture of examples, this discrepancy grows with the mean annotation entropy of the test set.  $\square$

**Remark 1** (Independence assumption). *Assumption (i) (that  $\hat{p}_{\hat{c}}(x)$  is approximately independent of  $H(x)$ ) may be violated in practice if harder examples (high  $H(x)$ ) also have lower model confidence. When the two are positively correlated, the per-example discrepancy term  $\delta(x)$  grows even faster with  $H(x)$  than the proof suggests, so Proposition 2 remains valid and the monotonicity is if anything conservative.*



## D Monte Carlo Temperature Scaling (MCTS)

**Method.** When individual annotation records are available rather than pre-aggregated label distributions, **MCTS** draws  $S$  samples  $a_{is} \sim \hat{\pi}(x_i)$  per example and minimizes

$$\mathcal{L}_{\text{MCTS}}(T) = -\frac{1}{nS} \sum_{i=1}^n \sum_{s=1}^S \log \text{softmax}(z_i/T)_{a_{is}}.$$

MCTS is a strictly more general formulation than SLTS: it applies even when only raw annotation samples are stored rather than pre-computed frequency vectors.

**A single annotation suffices.** The key practical implication of MCTS is that *even*  $S = 1$  — a single randomly drawn annotation per calibration example — already achieves  $\text{ECE}_{\text{true}}$  of 1.53%, matching SLTS to within rounding error and reducing TS’s 4.29% by 64%. This means that any dataset where each example has been annotated even once by a randomly selected annotator (rather than deterministically assigned the majority label) is immediately amenable to strong ambiguity-aware calibration. Formally, since  $\mathbb{E}_{\hat{y} \sim \hat{\pi}}[\text{CE}(\text{softmax}(z/T), \hat{y})] = \text{KL}(\hat{\pi} \parallel \text{softmax}(z/T)) + H(\hat{\pi})$ , the MCTS objective converges to SLTS in expectation as  $S \rightarrow \infty$ , with variance  $\propto 1/S$ . Table 3 confirms the fast convergence: the ECE gap between  $S = 1$  and  $S \rightarrow \infty$  is only 0.02 pp; in practice  $S = 20$ –50 is sufficient to reduce variance to negligible levels.

Table 3: **MCTS convergence on CIFAR-10H (ResNet-50).** Mean  $\pm$  std over 5 seeds;  $S$  = MC annotation samples per calibration example.

| Method                          | ECE (%)         | $T^*$             |
|---------------------------------|-----------------|-------------------|
| TS (voted labels)               | 4.29            | 2.030             |
| MCTS $S = 1$                    | $1.53 \pm 0.01$ | $3.174 \pm 0.041$ |
| MCTS $S = 5$                    | $1.53 \pm 0.02$ | $3.166 \pm 0.014$ |
| MCTS $S = 20$                   | $1.52 \pm 0.00$ | $3.187 \pm 0.013$ |
| MCTS $S = 50$                   | $1.52 \pm 0.01$ | $3.185 \pm 0.007$ |
| MCTS $S = 200$                  | $1.52 \pm 0.00$ | $3.178 \pm 0.003$ |
| SLTS ( $S \rightarrow \infty$ ) | 1.51            | 3.180             |

## 474 E Ablation: Calibration Set Size

475 We fix the annotation count at the full observed value and vary the fraction of calibration examples  
 476 used, from 5% to 100% (stratified subsampling by class). Results are shown for all four benchmarks  
 477 and eight architectures.

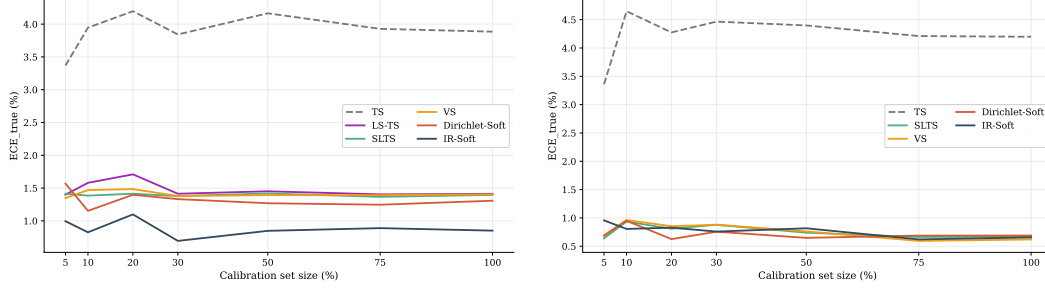


Figure 6: **Calibration-set-size ablation — CIFAR-10H.** Left: ResNet-50. Right: ViT-B/16. TS’s ECE is flat across all calibration set sizes; ambiguity-aware methods converge within 5–10%.

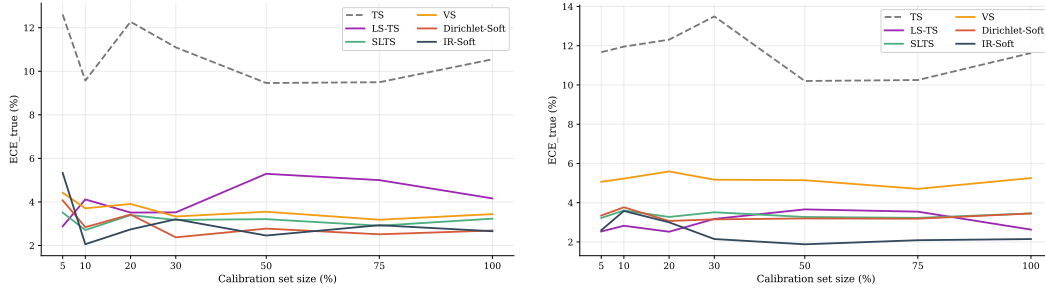


Figure 7: **Calibration-set-size ablation — ChaosNLI.** Left: RoBERTa-Large. Right: DeBERTa-v3-base. NLI models show the same pattern: TS is flat near 10–13% while soft methods stabilise near 3–4%.

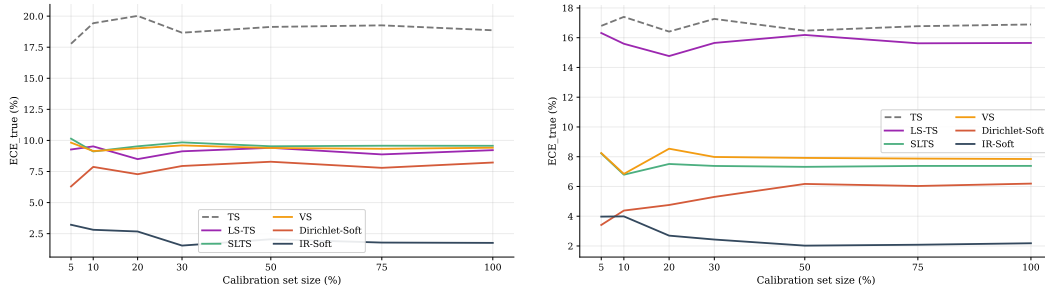


Figure 8: **Calibration-set-size ablation — ISIC 2019.** Left: EfficientNet-B4. Right: ViT-S/16. IR-Soft achieves the lowest ECE across all calibration set sizes on both architectures.

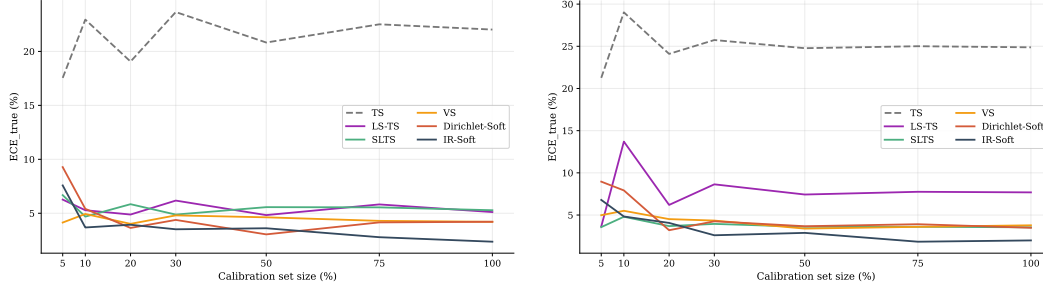


Figure 9: **Calibration-set-size ablation — DermaMNIST.** Left: ResNet-18. Right: ViT-S/16. Ambiguity-aware methods converge quickly; TS remains flat regardless of calibration set size.

478 **Key findings.** Across all four benchmarks and eight architectures, TS’s ECE does not decrease with  
 479 more calibration data, confirming that the degradation is a target bias rather than a variance artifact.  
 480 In contrast, SLTS achieves stable low ECE from as few as 5–10% of the calibration set, and IR-Soft  
 481 and Dirichlet-Soft converge even faster on most benchmarks. This suggests that the annotation  
 482 distribution (not the number of calibration examples) is the limiting factor for ambiguity-aware  
 483 methods.

## 484 F Ablation: Number of Annotations per Image

485 We study how performance depends on the number of annotations  $m$  per calibration image. We fix  
 486 the calibration set size and the annotation generation seed, and sweep  $m$  from 1 to 50, resampling the  
 487 soft targets for each value of  $m$ . Results are shown for CIFAR-10H, which provides up to  $\approx 51$  real  
 488 annotations per image; we subsample from the full pool.

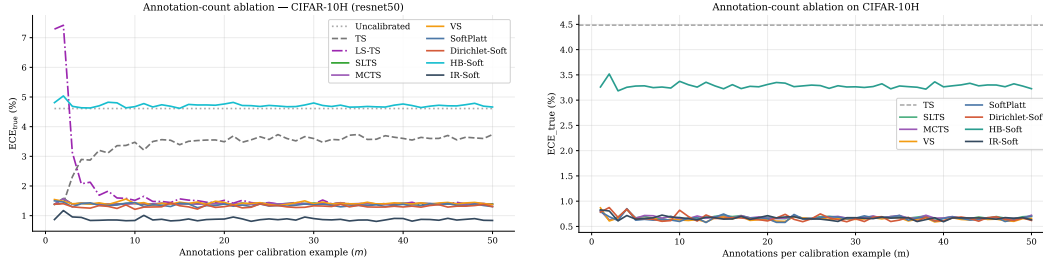


Figure 10: **Annotation-count ablation — CIFAR-10H.** Left: ResNet-50. Right: ViT-B/16.  $ECE_{\text{true}}$  as a function of annotations per image  $m$ . Ambiguity-aware methods (SLTS, Dirichlet-Soft, IR-Soft) are stable across nearly all  $m \geq 2$ , while TS rises steadily because its scalar temperature adjusts to the increasingly diffuse soft targets, degrading  $ECE_{\text{true}}$ . LS-TS starts high at  $m = 1$  (annotation noise dominates the smoothing estimate) but rapidly converges to  $\approx 1.4\%$  for  $m \geq 4$ .

489 **Key findings.** Ambiguity-aware calibration is robust to annotation count once  $m \geq 2$ : SLTS,  
 490 Dirichlet-Soft, and IR-Soft achieve near-minimal ECE with only a handful of annotations per image  
 491 and do not improve substantially beyond  $m = 5$ . LS-TS converges quickly (by  $m \approx 4$ ) because the  
 492 global smoothing parameter  $\varepsilon$  stabilises once each image has at least a few annotations. Standard  
 493 Temperature Scaling shows the opposite trend: its  $ECE_{\text{true}}$  *increases* with  $m$  because a higher  
 494 temperature is required to match the more diffuse annotation distribution, exposing the fundamental  
 495 target mismatch.

## 496 G Statistical Significance: Multi-Seed Analysis

497 For CIFAR-10H and ChaosNLI we repeat the evaluation with five random seeds (42–46), each  
 498 producing a different stratified 50/50 cal/test split. For ISIC 2019 and DermaMNIST the val/test split  
 499 is predetermined, so we instead vary the annotation generation seed (42–46), testing sensitivity to the  
 500 particular synthetic annotation sample. Table 4 reports mean  $\pm$  std of  $ECE_{\text{true}}$  across seeds for all  
 501 eight (dataset, arch) combinations.

Table 4: **Multi-seed stability — all benchmarks and architectures.** Mean  $\pm$  std of  $ECE_{\text{true}}$  (%) over 5 seeds. For CIFAR-10H and ChaosNLI the seed varies the cal/test split; for ISIC 2019 and DermaMNIST it varies the synthetic annotation sample. Standard deviations are small relative to the ECE gaps in all cases.

| Dataset    | Arch       | TS               | SLTS            | Dir.-Soft       | IR-Soft         |
|------------|------------|------------------|-----------------|-----------------|-----------------|
| CIFAR-10H  | ResNet-50  | $4.25 \pm 0.25$  | $1.61 \pm 0.12$ | $1.37 \pm 0.09$ | $0.89 \pm 0.12$ |
| CIFAR-10H  | ViT-B/16   | $4.39 \pm 0.20$  | $0.83 \pm 0.14$ | $0.74 \pm 0.07$ | $0.88 \pm 0.16$ |
| ChaosNLI   | RoBERTa-L  | $11.54 \pm 0.54$ | $3.85 \pm 0.34$ | $2.90 \pm 0.14$ | $2.40 \pm 0.18$ |
| ChaosNLI   | DeBERTa-v3 | $12.73 \pm 0.60$ | $4.41 \pm 0.54$ | $3.80 \pm 0.35$ | $2.61 \pm 0.41$ |
| ISIC 2019  | ENet-B4    | $18.84 \pm 0.17$ | $9.80 \pm 0.22$ | $8.45 \pm 0.20$ | $1.79 \pm 0.16$ |
| ISIC 2019  | ViT-S/16   | $16.93 \pm 0.16$ | $7.30 \pm 0.13$ | $6.20 \pm 0.19$ | $2.05 \pm 0.13$ |
| DermaMNIST | ResNet-18  | $22.43 \pm 0.28$ | $5.08 \pm 0.25$ | $3.53 \pm 0.39$ | $2.33 \pm 0.17$ |
| DermaMNIST | ViT-S/16   | $24.81 \pm 0.08$ | $3.61 \pm 0.28$ | $3.11 \pm 0.21$ | $2.09 \pm 0.14$ |

502 Across all benchmarks and architectures, standard deviations are small relative to the ECE gaps: TS  
 503 vs. ambiguity-aware methods differ by 2–10 $\times$  the standard deviation, confirming statistical robustness  
 504 of the results in Tables 1–2.

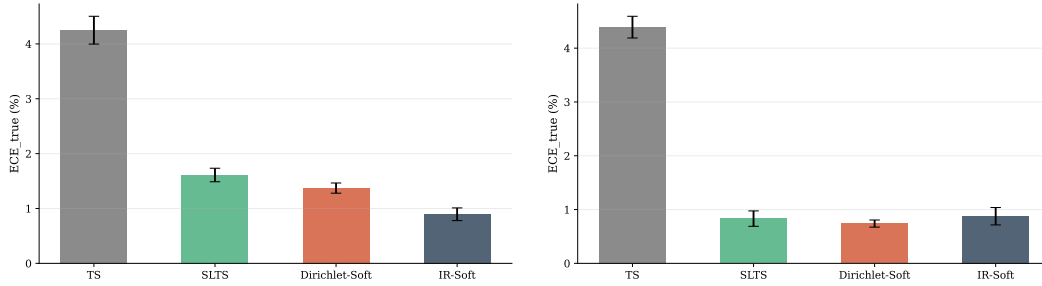


Figure 11: **Multi-seed bar charts — CIFAR-10H.** Left: ResNet-50. Right: ViT-B/16.

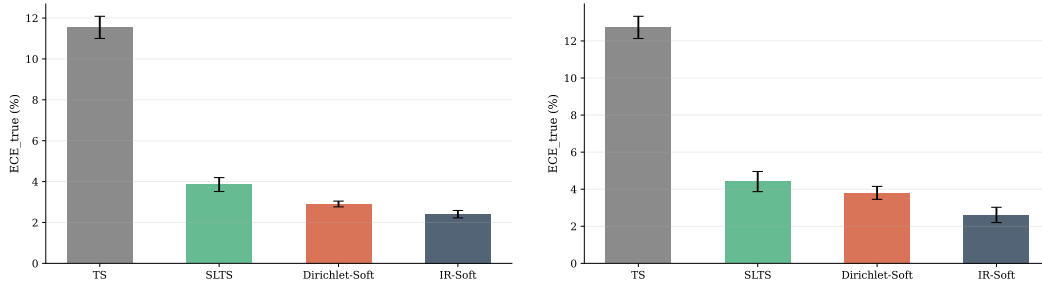


Figure 12: **Multi-seed bar charts — ChaosNLI.** Left: RoBERTa-Large. Right: DeBERTa-v3-base.

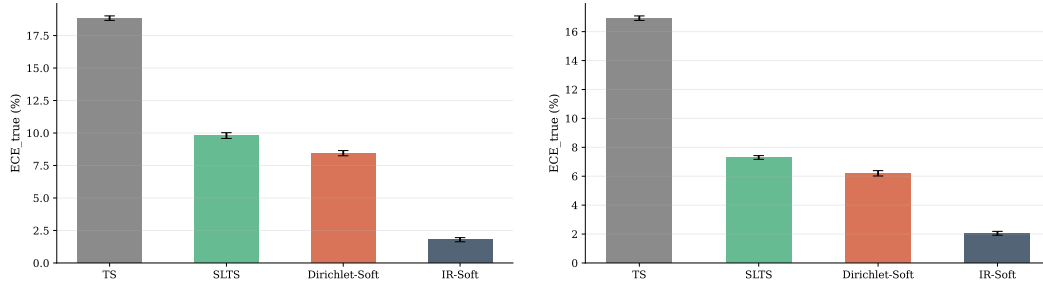


Figure 13: **Multi-seed bar charts — ISIC 2019.** Left: EfficientNet-B4. Right: ViT-S/16. Seed varies the annotation generation; val/test split is fixed.

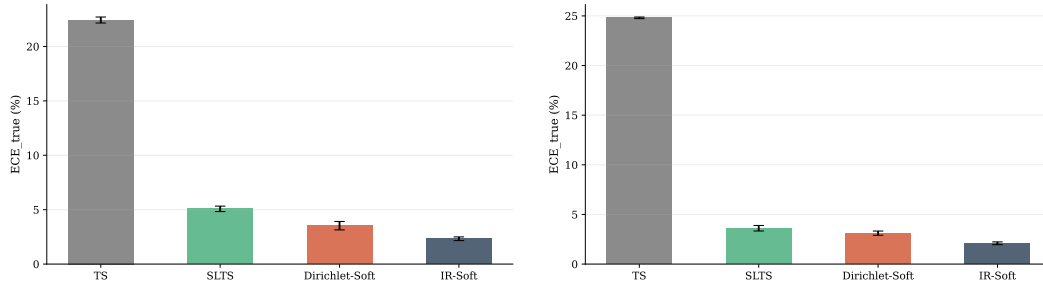


Figure 14: **Multi-seed bar charts — DermaMNIST.** Left: ResNet-18. Right: ViT-S/16. Seed varies the annotation generation; val/test split is fixed.

## 505 **H Reliability Diagrams**

506 Four-method summaries for CIFAR-10H (ResNet-50) and ChaosNLI (RoBERTa-Large) are shown in  
507 Figures 3 and 4 in the main text. Here we provide the four-method summary panels and full  $3 \times 5$   
508 panels for all remaining architectures and datasets. All panels share the same layout: top row = voted-  
509 label baselines (Uncal, TS, Platt, Dir.-Hard, HB-Hard); middle row = annotation-free (LS-TS) and  
510 temperature-based soft methods (SLTS, MCTS, SoftPlatt, VS); bottom row = non-parametric/matrix  
511 soft methods (Dir.-Soft, HB-Soft, IR-Soft). Red shading = overconfident; green = underconfident.

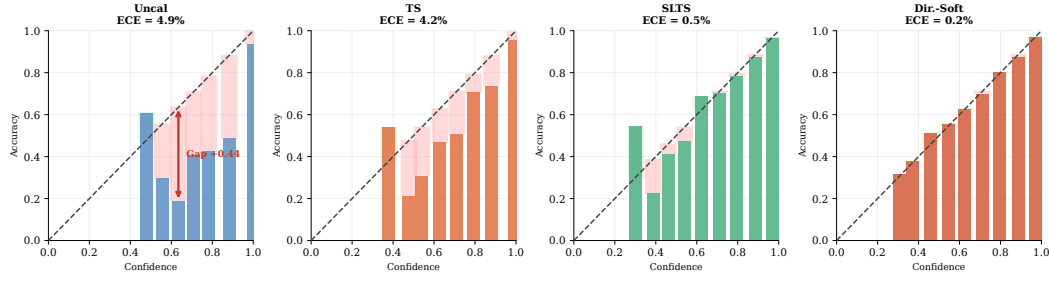
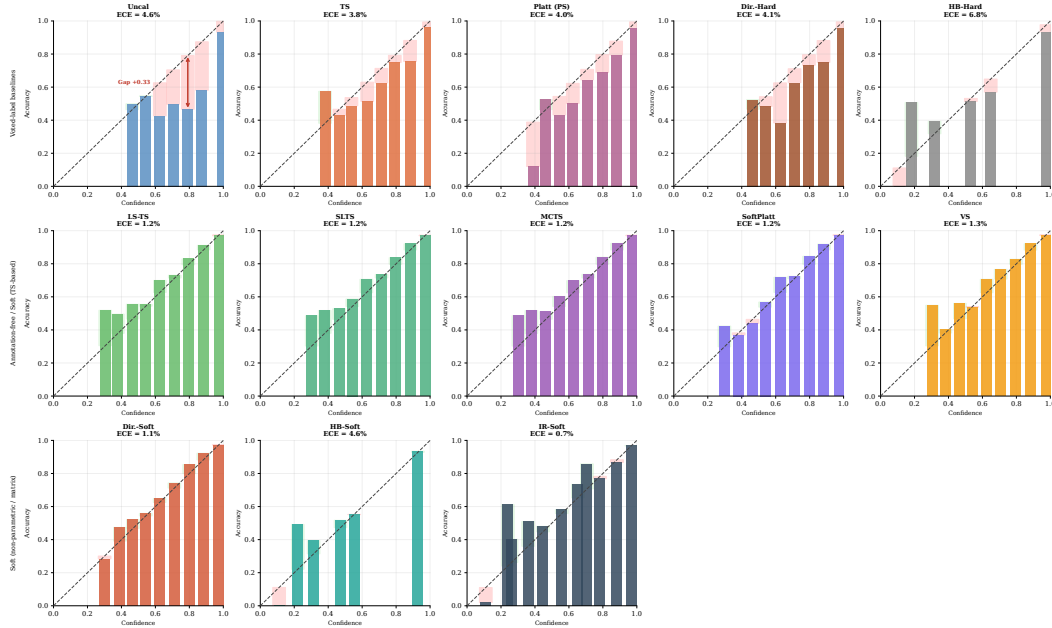


Figure 15: Reliability diagrams (summary) — CIFAR-10H ViT-B/16.

Figure 16: Reliability diagrams (all methods) — CIFAR-10H ResNet-50 ( $ECE_{true}$ ).



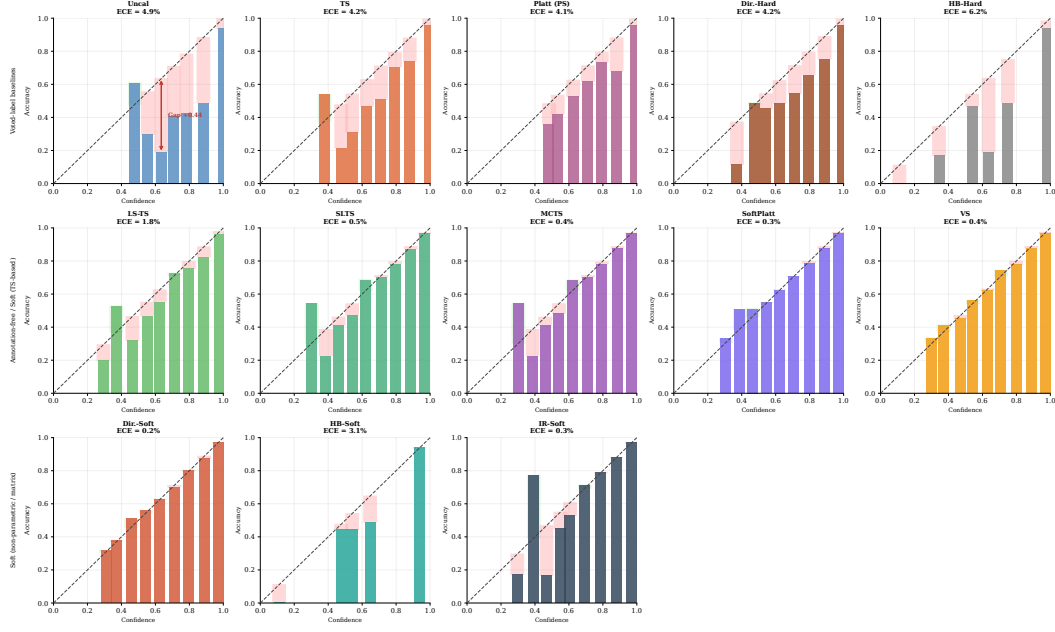


Figure 17: Reliability diagrams (all methods) — CIFAR-10H ViT-B/16 ( $ECE_{\text{true}}$ ).

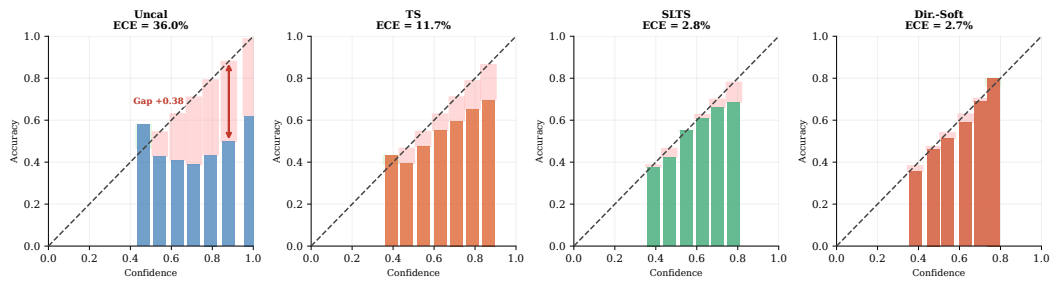


Figure 18: Reliability diagrams (summary) — ChaosNLI DeBERTa-v3-base.

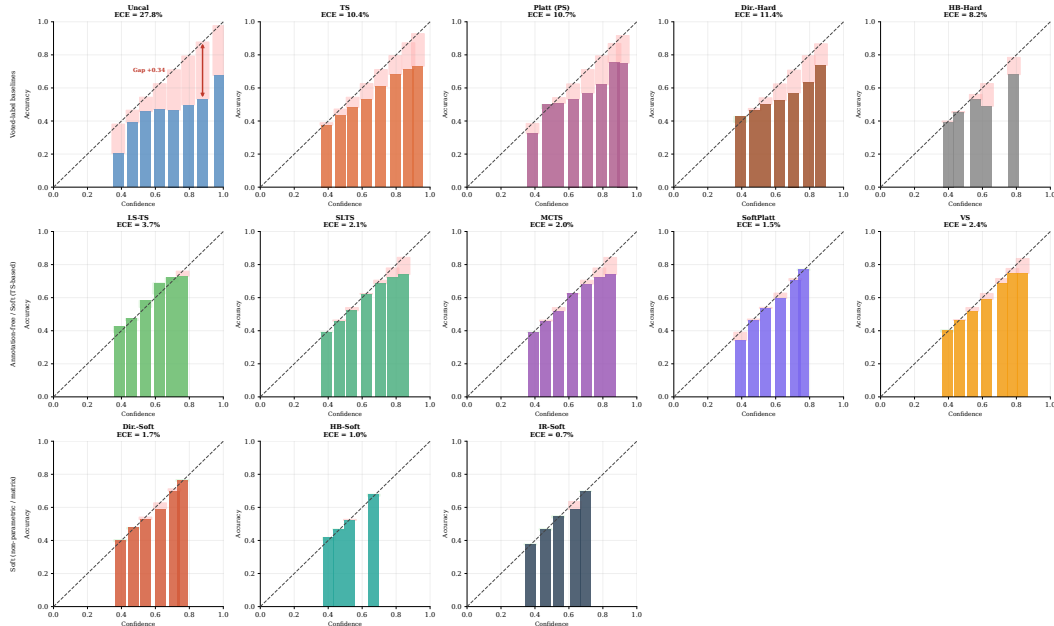


Figure 19: Reliability diagrams (all methods) — ChaosNLI RoBERTa-Large (ECE<sub>true</sub>). NLI models are severely overconfident before calibration; ambiguity-aware methods substantially correct this.

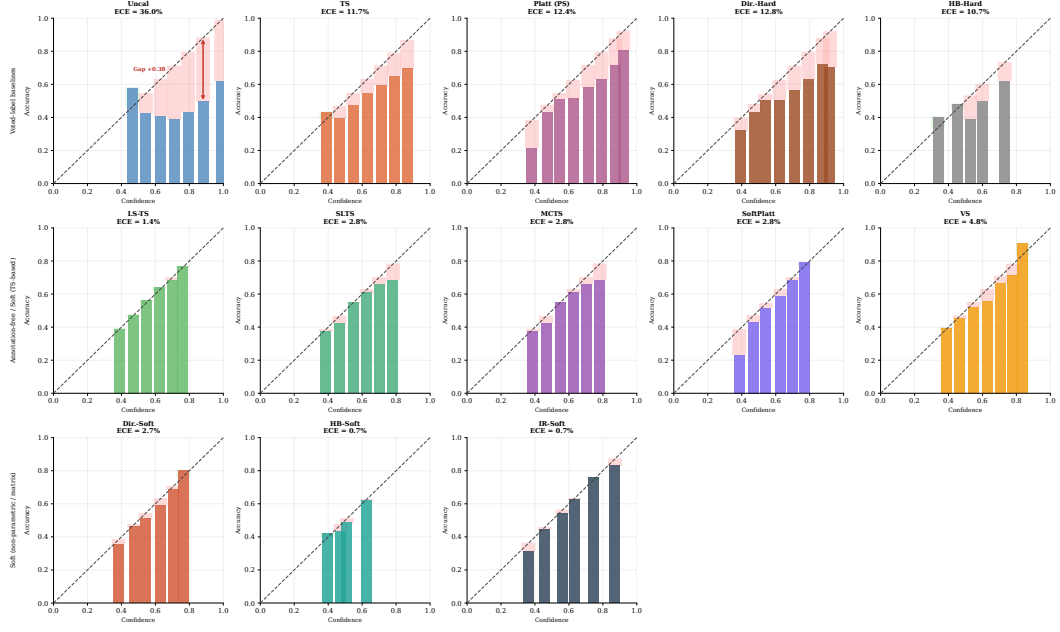


Figure 20: Reliability diagrams (all methods) — ChaosNLI DeBERTa-v3-base ( $ECE_{true}$ ).

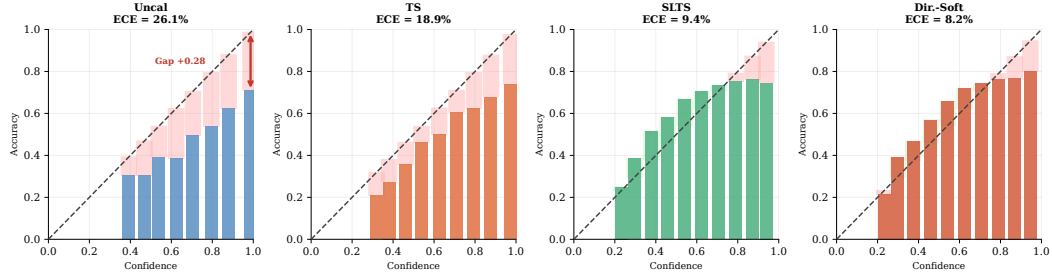


Figure 21: Reliability diagrams (summary) — ISIC 2019 EfficientNet-B4.

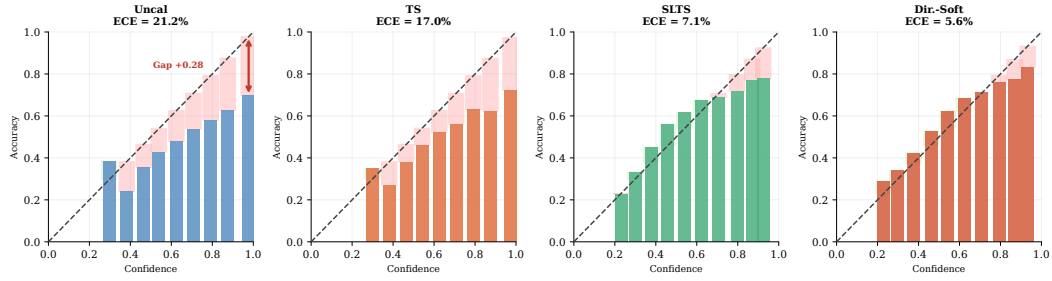
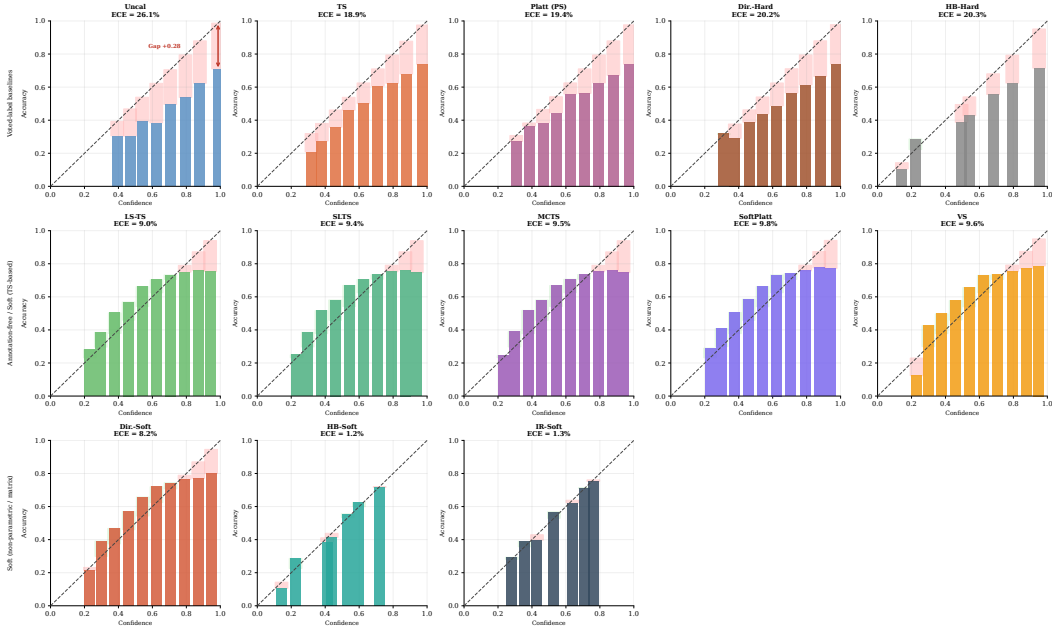


Figure 22: Reliability diagrams (summary) — ISIC 2019 ViT-S/16.

Figure 23: Reliability diagrams (all methods) — ISIC 2019 EfficientNet-B4 ( $ECE_{true}$ ).

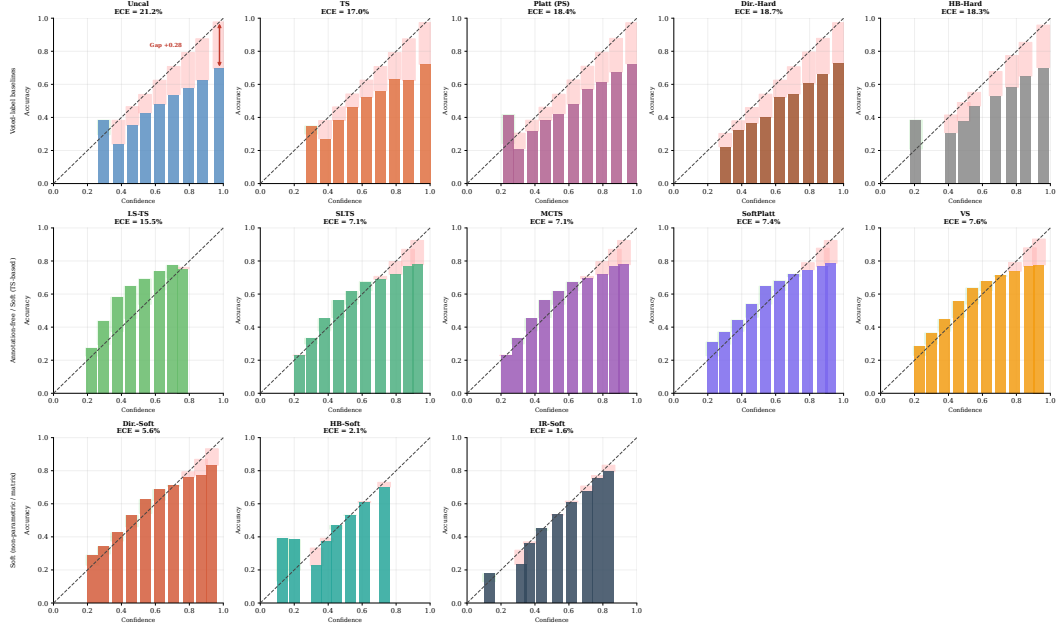


Figure 24: Reliability diagrams (all methods) — ISIC 2019 ViT-S/16 ( $ECE_{\text{true}}$ ).

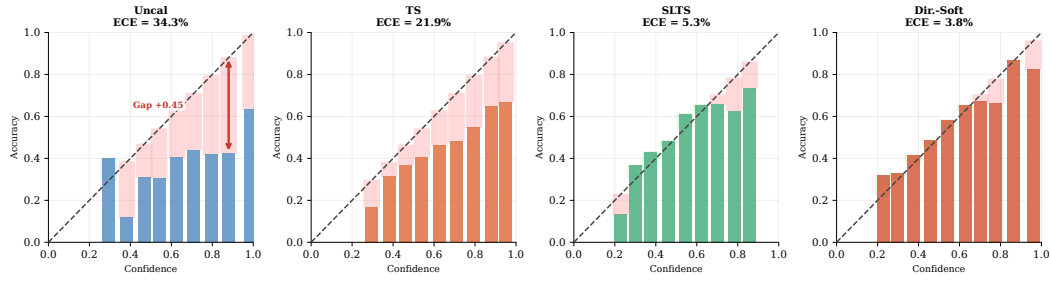


Figure 25: Reliability diagrams (summary) — DermaMNIST ResNet-18.

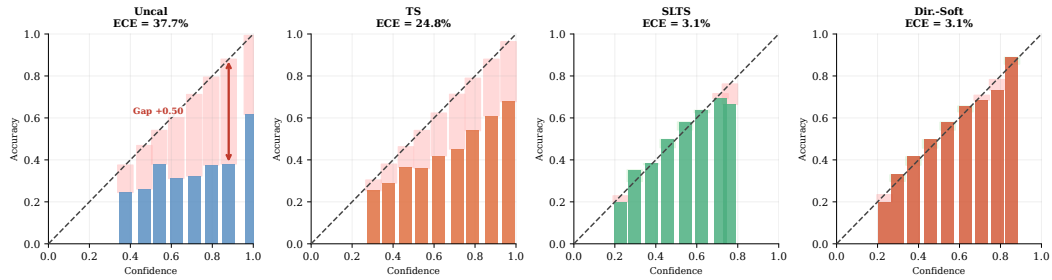
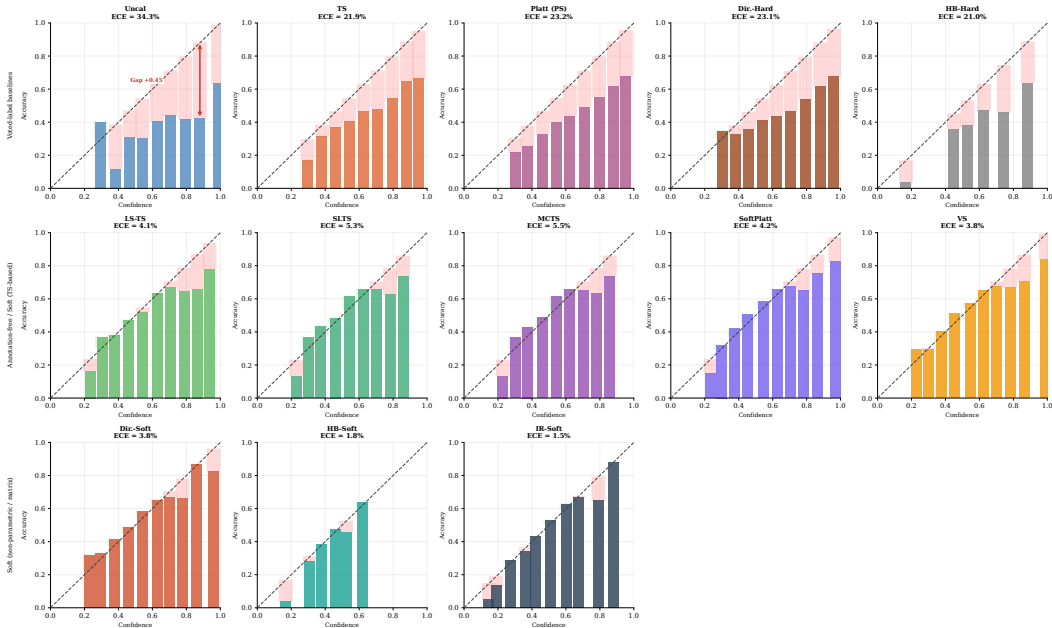


Figure 26: Reliability diagrams (summary) — DermaMNIST ViT-S/16.

Figure 27: Reliability diagrams (all methods) — DermaMNIST ResNet-18 (ECE<sub>true</sub>).

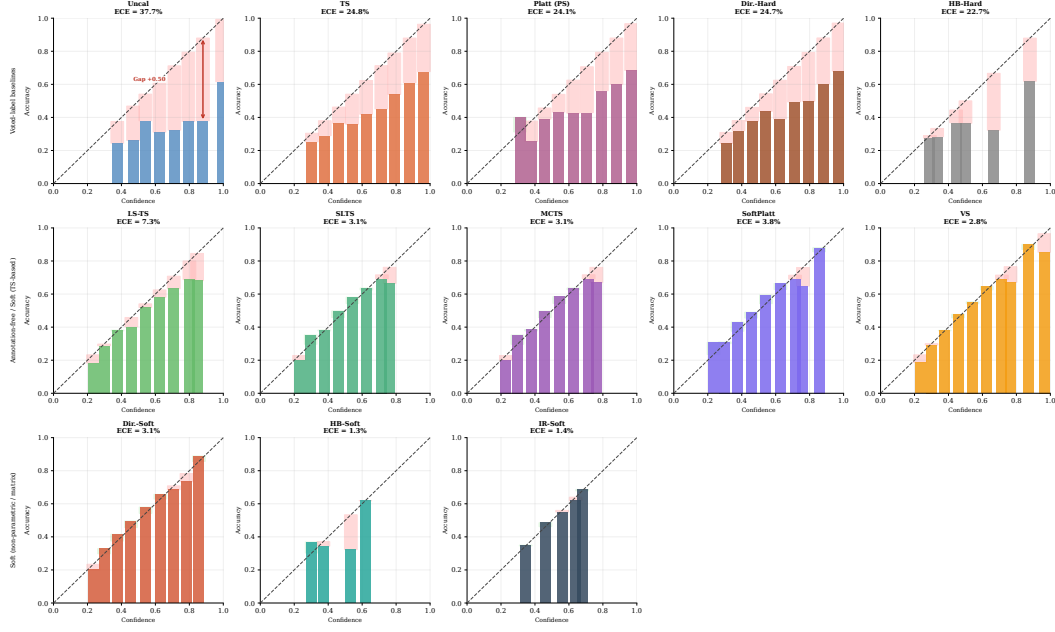


Figure 28: Reliability diagrams (all methods) — DermaMNIST ViT-S/16 ( $ECE_{true}$ ).

516 The diagrams confirm the quantitative results in Tables 1–2: TS reduces overconfidence relative to  
 517 *Uncal* but leaves a large residual gap because it targets voted labels. Dirichlet-Hard performs similarly  
 518 to TS and sometimes worse, confirming that the problem lies in the target, not model capacity. All  
 519 soft methods substantially close the gap, with IR-Soft and Dir.-Soft achieving the most uniform  
 520 residuals across all benchmarks.

## 521 I ISIC 2019 Annotator Confusion Matrix

### 522 Construction

523 Because the full per-image reader annotations of Liu et al. [2020] are not publicly available, we  
 524 construct a synthetic annotator model using a clinically calibrated  $8 \times 8$  confusion matrix  $C$ , where  
 525 entry  $C_{ij}$  gives the probability that a board-certified dermatologist labels an image as class  $j$  when  
 526 the consensus (majority-vote) diagnosis is class  $i$ .

527 **Calibration sources.** Diagonal entries (per-condition agreement rates) are set to match the per-  
 528 condition inter-reader agreement reported in Liu et al. [2020] (Table 2, nine board-certified dermatol-  
 529 ogists reading 963 clinical images) and corroborated by Haenssle et al. [2018]. Off-diagonal entries  
 530 encode clinically established confusion patterns:

- 531 • **MEL/NV** (Melanoma / Melanocytic Nevi): the most consequential and most confused pair  
 532 in dermoscopy, with diagonal rates of 73%/76%. Off-diagonal entries  $C_{\text{MEL},\text{NV}} = 0.14$  and  
 533  $C_{\text{NV},\text{MEL}} = 0.15$  reflect the well-documented difficulty of distinguishing early melanoma  
 534 from benign nevi [Haenssle et al., 2018].
- 535 • **AK/BKL/SCC** (Actinic Keratosis / Benign Keratosis-like Lesions / Squamous Cell Car-  
 536 cinoma): a high-confusion cluster with diagonal rates of 65%/62%/67%. The  $\text{AK} \leftrightarrow \text{SCC}$   
 537 confusion ( $C_{\text{AK},\text{SCC}} = 0.16$ ,  $C_{\text{SCC},\text{AK}} = 0.18$ ) is clinically significant because untreated AK  
 538 can progress to SCC. BKL shares morphological features with both conditions.
- 539 • **DF/VL** (Dermatofibroma / Vascular Lesion): the easiest to distinguish, with diagonal rates  
 540 of 87%/91%.

541 The mean diagonal agreement is  $\bar{C}_{ii} = 75\%$ , matching the aggregate inter-reader agreement reported  
 542 by Liu et al. [2020] exactly.

543 **Simulation procedure.** For each test image  $x_i$  with consensus label  $y_i$ , we independently draw  
 544  $m = 9$  synthetic annotations from the categorical distribution defined by row  $y_i$  of  $C$ :

$$a_i^{(k)} \sim \text{Categorical}(C_{y_i, \cdot}), \quad k = 1, \dots, m,$$

545 and set the empirical label distribution to:  $\pi(x_i) = \frac{1}{m} \sum_{k=1}^m \mathbf{e}_{a_i^{(k)}}$ . This directly mirrors the  $m = 9$   
 546 board-certified dermatologist annotation protocol of Liu et al. [2020], whose reader study we cannot  
 547 fully replicate due to data unavailability. The resulting mean annotation entropy over the ISIC 2019  
 548 test set is  $\bar{H} = 0.66$ , reflecting the genuine diagnostic difficulty of the 8-class task.

### 549 Visualisation

550 Figure 29 shows the confusion matrix as a heatmap. The MEL/NV cluster (orange dashed box) and  
 551 the AK/BKL/SCC high-confusion cluster (purple dotted highlights) are visually prominent. DF and  
 552 VL have near-perfect diagonal entries and negligible off-diagonal mass.



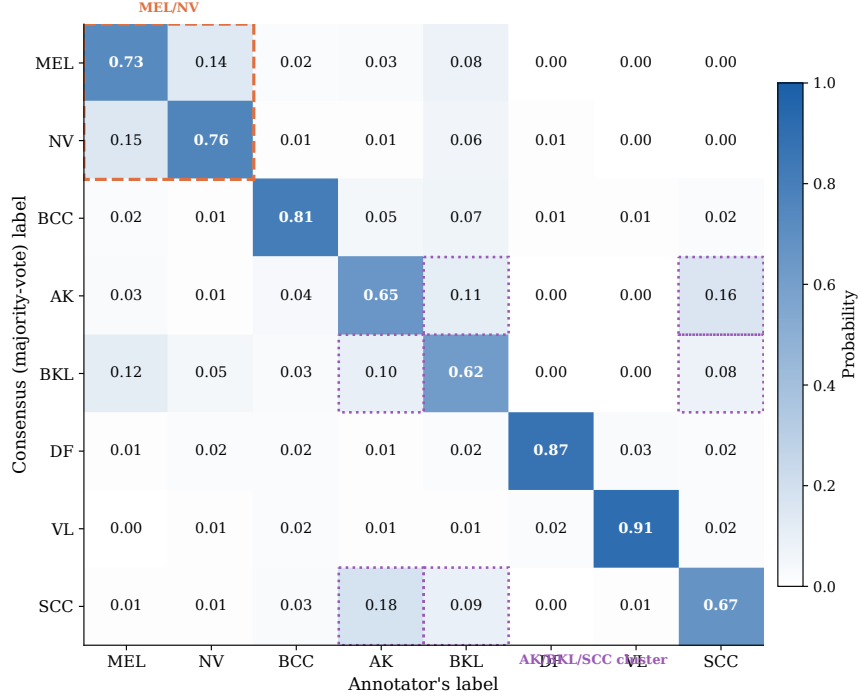


Figure 29: **Inter-reader confusion matrix for ISIC 2019.** Entry  $C_{ij}$  is the probability that a dermatologist labels an image as class  $j$  given the consensus label is  $i$ . Diagonal entries (bold) are the per-condition agreement rates, calibrated to match Liu et al. [2020] Table 2. Orange dashed box: MEL/NV most-confused pair. Purple dotted highlights: AK/BKL/SCC high-confusion cluster. Mean diagonal agreement  $\bar{C}_{ii} = 75\%$ .

### 553 Full confusion matrix

554 Table 5 lists the exact numerical entries of the confusion matrix.

Table 5: **Numerical values of the inter-reader confusion matrix for ISIC 2019.** Row = consensus (majority-vote) label; column = annotator’s label. Each row sums to 1. Diagonal entries (bold) are per-condition agreement rates, calibrated to Liu et al. [2020] Table 2; mean diagonal agreement  $\bar{C}_{ii} = 75\%$ .

|     | MEL         | NV          | BCC         | AK          | BKL         | DF          | VL          | SCC         |
|-----|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| MEL | <b>0.73</b> | 0.14        | 0.02        | 0.03        | 0.08        | 0.00        | 0.00        | 0.00        |
| NV  | 0.15        | <b>0.76</b> | 0.01        | 0.01        | 0.06        | 0.01        | 0.00        | 0.00        |
| BCC | 0.02        | 0.01        | <b>0.81</b> | 0.05        | 0.07        | 0.01        | 0.01        | 0.02        |
| AK  | 0.03        | 0.01        | 0.04        | <b>0.65</b> | 0.11        | 0.00        | 0.00        | 0.16        |
| BKL | 0.12        | 0.05        | 0.03        | 0.10        | <b>0.62</b> | 0.00        | 0.00        | 0.08        |
| DF  | 0.01        | 0.02        | 0.02        | 0.01        | 0.02        | <b>0.87</b> | 0.03        | 0.02        |
| VL  | 0.00        | 0.01        | 0.02        | 0.01        | 0.01        | 0.02        | <b>0.91</b> | 0.02        |
| SCC | 0.01        | 0.01        | 0.03        | 0.18        | 0.09        | 0.00        | 0.01        | <b>0.67</b> |

## 555 J Implementation Details

556 **CIFAR-10H models.** *ResNet-50*: ImageNet-1k weights (`ResNet50_Weights.IMAGENET1K_V1`);  
557 FC layer replaced with  $2048 \times 10$  linear; AdamW, lr=  $10^{-4}$ , weight decay  $10^{-4}$ , batch 128, cosine  
558 annealing, 30 epochs,  $224 \times 224$ . *ViT-B/16*: ImageNet-21k weights; classification head replaced;  
559 AdamW, lr=  $5 \times 10^{-5}$ , weight decay  $10^{-4}$ , batch 64, cosine annealing, 20 epochs,  $224 \times 224$ .

560 **ChaosNLI models.** *RoBERTa-Large*: `roberta-large-mnli` from HuggingFace, pre-fine-  
561 tuned on MultiNLI [Liu et al., 2019]; no additional fine-tuning. *DeBERTa-v3-base*:  
562 `cross-encoder/nli-deberta-v3-base` from HuggingFace [He et al., 2023]; no additional fine-  
563 tuning. Both models use the combined ChaosNLI SNLI+MNLI subset ( $n = 3,113$ ), split 50/50 into  
564 calibration and test (stratified, seed 42).

565 **ISIC 2019 models.** *EfficientNet-B4*: ImageNet-1k weights; stratified 70/15/15 split; class-weighted  
566 cross-entropy (NV:  $\approx 67\%$  of training images); AdamW, lr=  $3 \times 10^{-4}$ , weight decay  $10^{-4}$ , 2-epoch  
567 linear warm-up + cosine decay, 20 epochs,  $380 \times 380$ . *ViT-S/16*: ImageNet-21k weights; same split  
568 and loss weighting; AdamW, lr=  $10^{-4}$ , weight decay  $10^{-4}$ , 2-epoch warm-up + cosine decay, 20  
569 epochs,  $224 \times 224$ .

570 **DermaMNIST models.** *ResNet-18*: ImageNet-1k weights, class-weighted cross-entropy, AdamW,  
571 lr=  $10^{-4}$ , 30 epochs. *ViT-S/16*: ImageNet-21k weights, same training protocol,  $224 \times 224$ .

572 **Calibration optimizers.** TS, SLTS: LBFGS, lr= 0.1, 500 iterations, tolerance  $10^{-9}$  (gradient),  
573  $10^{-11}$  (loss). Platt/VS: Adam, lr= 0.01/0.05, weight decay  $10^{-4}$ , 2000 steps.

574 **ECE bins.**  $B = 15$  equal-width bins in  $[0, 1]$ ; bins with fewer than 1 example are excluded.

575 **Compute.** CIFAR-10H fine-tuning:  $\approx 20$  min on one NVIDIA A100. All calibration methods:  
576  $< 60$  s on CPU after logit caching.